IPSA – Institut Polytechnique des Sciences Avancées

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Homework of Flying and Handling Qualities

Mécanique du vol : Qualités de vol

Aircraft Model: C2

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Introduction

The study of aircraft dynamic stability is a fundamental discipline in flight mechanics and aeronautical engineering. It aims to characterize how an aircraft behaves when subjected to disturbances from its trimmed equilibrium flight condition, and to determine whether — and how quickly — it returns to that condition over time. This analysis is essential to ensure both the safety of the aircraft and the effectiveness of the pilot in accomplishing the intended mission.

The assessment of dynamic behavior is formalized through the concept of flying and handling qualities, which provide quantitative criteria linking the aircraft's dynamic characteristics to pilot workload and mission performance. These criteria are defined by the international military specification MIL-F-8785C, which classifies aircraft, flight phases and quality levels, and specifies requirements for each stability mode.

The objective of this report is to conduct a complete dynamic stability analysis of the C2 aircraft, a small, light airplane (Class I) operating in a cruise flight condition at an altitude of 7620 m and a true airspeed of 185.9 m/s (Mach 0.6), corresponding to Category B flight phase. The study is based on the linearized small perturbation theory applied around this steady trimmed condition.

The equations of motion are decoupled into two independent systems — longitudinal and lateral-directional dynamics — each represented in state-space form. All numerical computations are performed using MATLAB, from the construction of the state and control matrices to the eigenvalue analysis, time response simulations, transfer function extraction, Bode diagrams, Root Locus plots, and Stability Augmentation System design.

The report is organized as follows. Chapter 1 introduces the theoretical framework of flying and handling qualities, including the MIL-F-8785C classification and requirements. Chapter 2 develops the mathematical model of aircraft motion, from the general Newton's laws to the linearized state-space representation and the characterization of each stability mode. Chapter 3 applies this methodology to the C2 aircraft, presents all numerical results, assesses the flying quality level of each mode, and designs augmentation systems where appropriate.

Chapter 1: Flying Quality Requirements

Flying quality requirements provide a quantitative framework for assessing how easily and safely an aircraft can be controlled under various flight conditions. These requirements are essential in the aircraft design process, as they establish measurable criteria that link the dynamic characteristics of the aircraft to pilot workload and mission effectiveness. This chapter presents the fundamental definitions, classification systems, and specific criteria defined by the international military specification MIL-F-8785C, which remains the primary reference standard for flying qualities of piloted airplanes.

1.1 Definitions

1.1.1 Flying Qualities (FQ)

Flying Qualities are the stability and control characteristics of an aircraft that have an important effect on the safety of flight and on the pilot's impression of the ease of flying the airplane in steady flight and in maneuvers. They are the dynamic response characteristics of the aircraft that determine the ease (pilot workload) and precision of controlled flight. Flying qualities depend on the aircraft's aerodynamic and dynamic stability properties. They may be regarded as being task-related.

1.1.2 Handling Qualities (HQ)

Handling Qualities are those qualities or characteristics of an airplane that govern the ease and precision with which the pilot can perform the tasks required in support of the airplane's role. They represent the ability of the pilot to control the aircraft precisely and comfortably with minimal workload, and result from the combined effect of flying qualities, control system design, cockpit cues, and pilot-vehicle interaction. Handling qualities may be regarded as being response-related.

1.1.3 Aircraft Stability

Aircraft stability is the ability of an airplane to maintain or return to its original flight path after any disturbance encountered in flight. Two types of stability are distinguished:

> Static stability: the instantaneous tendency of the airplane to return to its initial position after being submitted to a perturbation. Three configurations exist: positive static stability (aircraft tends to return to original attitude), negative static stability (aircraft tends to deviate from original attitude), and neutral static stability (aircraft maintains its new attitude). Aircraft must possess positive static stability, which depends on aircraft design and trimming.
>
> Dynamic stability: describes how an airplane responds over time to a disturbance. A dynamically stable aircraft will see its oscillations decrease in amplitude over time until it returns to equilibrium.

Three kinds of positive static stability are considered for aircraft design: longitudinal static stability (response to pitch disturbance), lateral static stability (response to roll disturbance), and directional static stability (response to yaw disturbance).

1.1.4 Control

Aircraft control refers to the ability of the pilot to command changes in the aircraft's flight path and attitude through the use of control surfaces (elevator, ailerons, rudder) and thrust. Adequate control requires that the aircraft respond predictably and proportionally to pilot inputs across the entire flight envelope.

1.1.5 Pilot-In-the-Loop Oscillation (PIO)

A Pilot-In-the-Loop Oscillation (PIO) is an unintentional sustained or uncontrollable oscillation that results from the efforts of the pilot to control the aircraft. PIOs arise when there is an unfavorable coupling between the pilot's control inputs and the aircraft's dynamic response, often due to time delays or inadequate damping. Designing aircraft for satisfactory flying qualities aims to avoid PIOs.

1.1.6 Relationship Between These Notions

Stability, control, and flying qualities are interconnected concepts. Stability provides the foundation upon which controllability is built: an aircraft must first possess adequate stability for the pilot to control it effectively. Flying qualities are then assessed by evaluating how the stability and control characteristics combine to affect pilot workload during specific mission tasks. Handling qualities represent the final pilot-centered assessment of the overall aircraft-pilot system. The goal of the flying quality requirements is to ensure that the aircraft possesses sufficient levels of stability and control to achieve satisfactory handling qualities across all flight phases.

1.2 Classification of Airplanes (MIL-F-8785C)

The military specification MIL-F-8785C classifies airplanes into four classes based on their size, weight, and maneuverability characteristics. This classification determines which set of flying quality requirements applies to a given aircraft.

Class	Description	Examples
I	Small, light airplanes	Utility aircraft, primary trainers, general aviation
II	Medium weight, low to medium maneuverability	Transport aircraft, tactical bombers
III	Large, heavy airplanes	Heavy transport, tankers, heavy bombers
IV	Highly maneuverable airplanes	Fighter aircraft, attack aircraft

1.3 Flight Phase Categories

MIL-F-8785C defines three flight phase categories that reflect the type of maneuvering expected during different segments of the mission. The applicable flying quality requirements depend on the flight phase category.

Category A: Non-terminal flight requiring rapid maneuvering

This category covers flight phases requiring precise tracking or precise flight path control. It includes:

- Air-to-air combat
- Ground attack
- In-flight refuelling (receiver)
- Close reconnaissance
- Terrain following
- Close formation flying

Category B: Non-terminal flight requiring gradual maneuvering

This category covers less demanding flight phases such as:

- Climb
- Cruise
- Descent
- In-flight refuelling (tanker)
- Loiter

Category C: Terminal flight phases

This category covers flight phases in the vicinity of the airfield:

- Take-off (normal and catapult)
- Approach
- Wave-off / Go-around
- Landing

1.4 Levels of Flying Qualities

Three levels of flying qualities are defined, reflecting the degree of pilot compensation required to maintain adequate aircraft control:

Level	Description
Level 1	Flying qualities clearly adequate for the mission flight phase. Desired performance is achievable with no more than minimal pilot compensation.
Level 2	Flying qualities adequate to accomplish the mission flight phase, but some increase in pilot workload or degradation in mission effectiveness, or both, exists.
Level 3	Flying qualities such that the aircraft can be controlled in the context of the mission flight phase, even though pilot workload is excessive or mission effectiveness is inadequate, or both.

The objective in aircraft design is to achieve Level 1 flying qualities for all normal flight phases. Level 2 may be acceptable for certain degraded conditions, while Level 3 represents the minimum acceptable standard ensuring the aircraft remains controllable.

The Cooper-Harper Handling Qualities Rating Scale (HQRS) provides a complementary subjective assessment tool. This scale ranges from 1 (excellent) to 10 (major deficiencies, loss of control), and is used by test pilots to evaluate handling qualities during flight testing. Ratings 1-3 correspond to Level 1, ratings 4-6 to Level 2, and ratings 7-9 to Level 3.

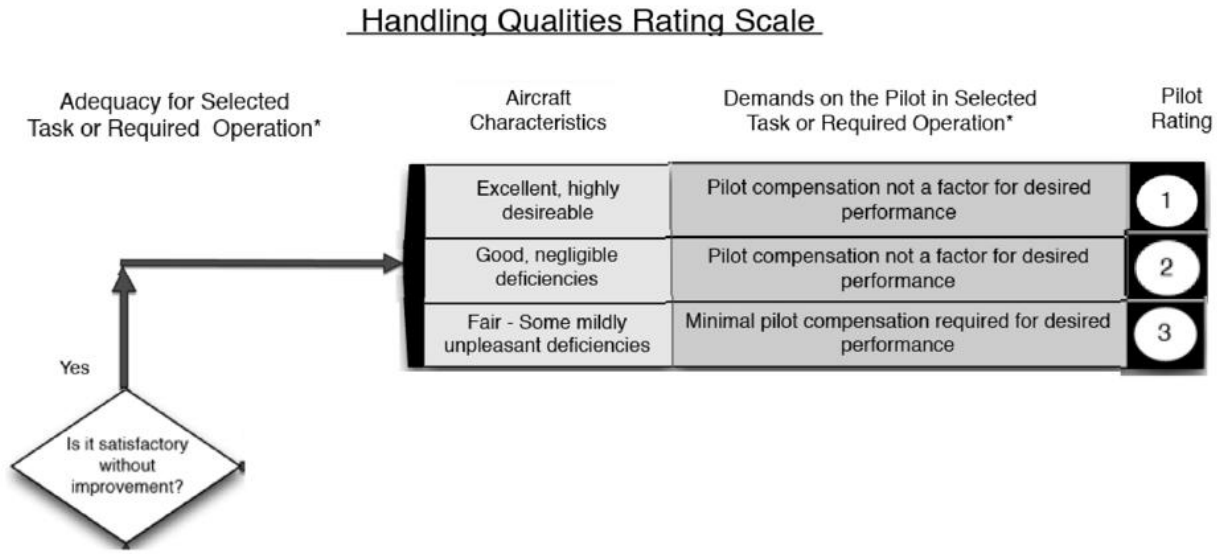


Figure 1.1: Cooper-Harper Handling Qualities Rating Scale (part 1 / 2).

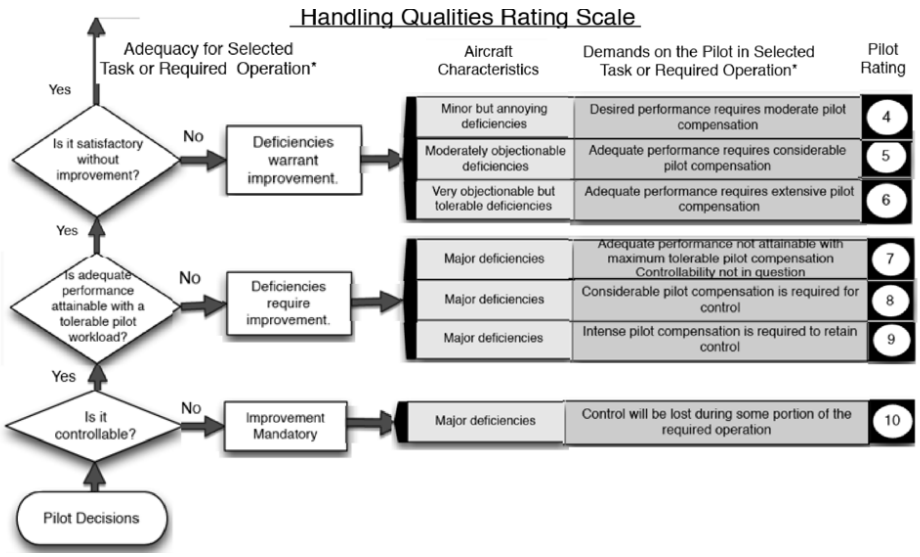


Figure 1.2: Cooper-Harper Handling Qualities Rating Scale (part 2 / 2, complete decision tree).

1.5 Longitudinal Flying Quality Requirements

Longitudinal flying quality requirements address the two longitudinal stability modes: the short period mode and the phugoid mode.

1.5.1 Short Period Mode Requirements

The short period mode is the high-frequency longitudinal oscillation that governs the transient incidence response after an elevator deflection. Its characteristics are critical for pilot perception of aircraft responsiveness. Three criteria are used to evaluate this mode.

Criterion 1: Short Period Damping Ratio Limits

The first criterion specifies acceptable ranges for the short period damping ratio ξ_SP as a function of flight phase category and flying quality level.

Flight Phase Category	Level 1 ξ_min	Level 1 ξ_max	Level 2 ξ_min	Level 2 ξ_max	Level 3 ξ_min	Level 3 ξ_max
Cat A	0.35	1.30	0.25	2.00	0.10	2.00
Cat B	0.30	2.00	0.20	2.00	0.10	2.00
Cat C	0.50	1.30	0.35	2.00	0.25	2.00

Table 1.1: Short period damping ratio limits per MIL-F-8785C

Criterion 2: Short Period Natural Frequency vs. Damping Ratio

The second criterion evaluates the short period mode on a diagram of natural frequency ω_n_SP versus damping ratio ξ_SP. This diagram defines regions of satisfactory, acceptable, poor, and

unacceptable flying qualities. The boundaries depend on the flight phase category and aircraft class.

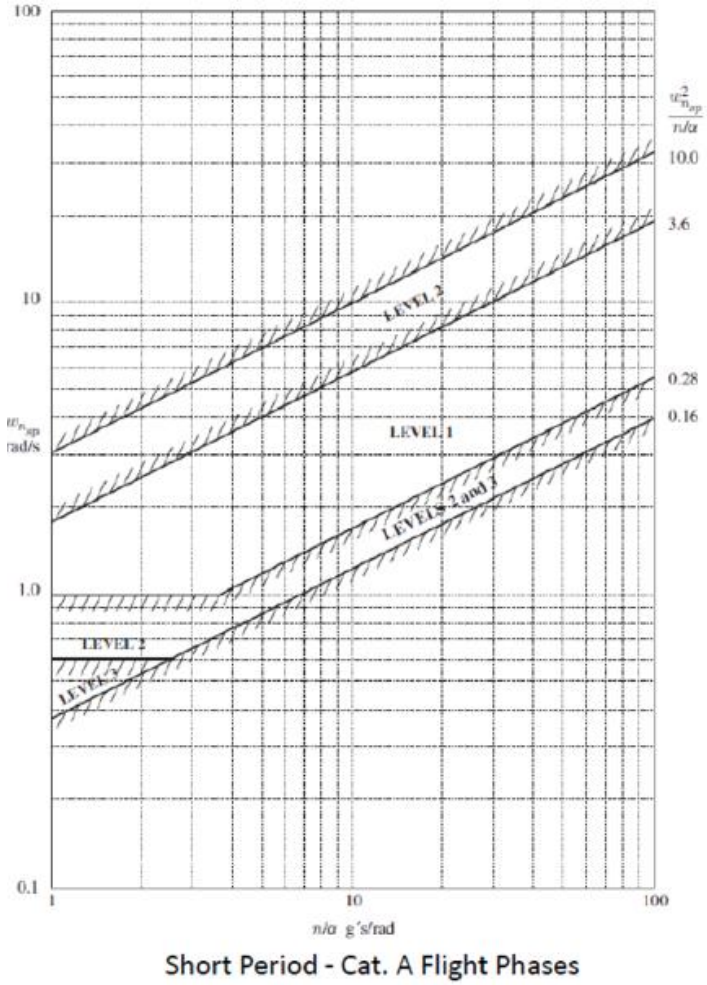


Figure 1.3: Short period natural frequency ω_n_SP versus damping ratio ξ_SP (MIL-F-8785C). Regions of Satisfactory, Acceptable, Poor and Unacceptable flying qualities are shown

Criterion 3: Control Anticipation Parameter (CAP)

The Control Anticipation Parameter (CAP) is the most discriminating criterion for short period flying qualities. It relates the initial pitch acceleration to the steady-state normal acceleration following a step elevator input. The CAP is defined as:

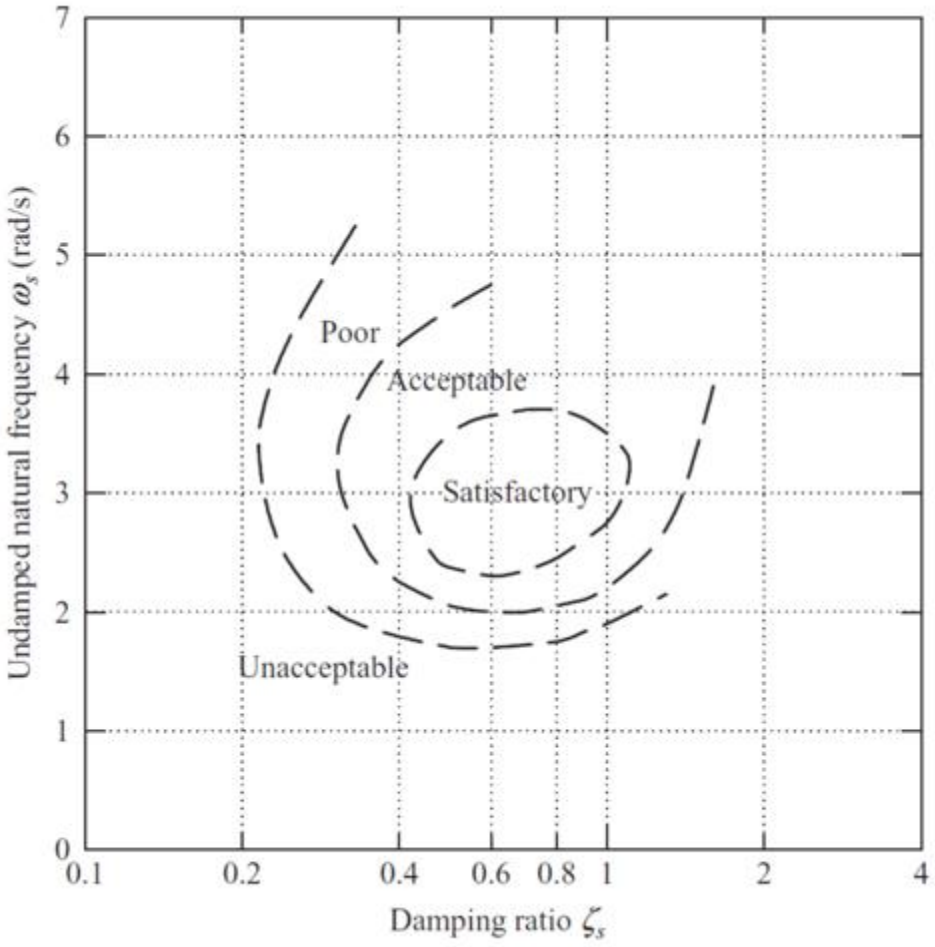


Figure 1.4: Control Anticipation Parameter (CAP) versus short period damping ratio ξ_SP — Flight Phase Category A. Level 1, 2 and 3 acceptable regions are shown as rectangular boxes per MIL-F-8785C.

$$CAP = \omega^2_n_SP \,/\, (n/\alpha)$$

where n/α is the acceleration sensitivity, defined as:

$$n/\alpha = (\bar{q} \cdot S \cdot C_L\alpha) \,/\, (m \cdot g)$$

with $\bar{q}$ the dynamic pressure, S the wing reference area, C_Lα the lift curve slope, m the aircraft mass, and g the gravitational acceleration.

Alternatively, the CAP can be computed from the dimensional stability derivatives as:

$$CAP = [M_q \cdot (-Z_w \,/\, U_0) + M_w] \,/\, (-Z_w \,/\, g)$$

The CAP is plotted against the short period damping ratio ξ_SP on a diagram that defines Level 1, Level 2, and Level 3 boundaries for each flight phase category.

1.5.2 Phugoid Mode Requirements

The phugoid is the low-frequency, long-period oscillation that primarily affects the aircraft speed and altitude at nearly constant angle of attack. Being a slow oscillation, it is generally well

controlled by the pilot. The requirements are therefore less stringent than for the short period mode:

Level	Requirement
Level 1	$\xi_{Ph} \geq 0.04$
Level 2	$\xi_{Ph} \geq 0$ (phugoid must not be unstable)
Level 3	$T_{2,Ph} \geq 55$ seconds

Table 1.2: Phugoid mode flying quality requirements

For Level 3, the requirement uses the time to double amplitude T_2, defined as:

$$T_{2,Ph} = \ln(2) / (-\xi_{Ph} \cdot \omega_{n,Ph}) = 0.693 / |\xi_{Ph} \cdot \omega_{n,Ph}|$$

This formula applies when the phugoid is unstable ($\xi_{Ph} < 0$). T_2 represents the time for the oscillation amplitude to double, and must be at least 55 seconds to give the pilot sufficient time to react.

1.6 Lateral Flying Quality Requirements

Lateral flying quality requirements address the three lateral-directional stability modes: the roll mode, the spiral mode, and the Dutch roll mode.

1.6.1 Roll Subsidence Mode Requirements

The roll mode is a non-oscillatory (exponential) mode governed by roll damping. The requirement specifies a maximum allowable roll mode time constant T_R to ensure the aircraft responds quickly to aileron inputs:

Flight Phase Category	Level 1 T_R max (s)	Level 2 T_R max (s)	Level 3 T_R max (s)
Cat A, C (Class I, IV)	1.0	1.4	10
Cat A, C (Class II, III)	1.4	3.0	10
Cat B (all classes)	1.4	3.0	10

Table 1.3: Maximum roll mode time constant per MIL-F-8785C

The roll mode time constant is computed as $T_R = -1/\lambda_{Roll}$, where λ_{Roll} is the roll mode eigenvalue (real, negative for a stable mode).

1.6.2 Spiral Mode Requirements

The spiral mode is a slow, non-oscillatory mode that manifests as a gradual engagement into a banked turn. A mildly unstable spiral mode is tolerable because the pilot can easily correct it. The requirement specifies a minimum time to double amplitude T_2 when the spiral mode is unstable:

Flight Phase Category	Level 1 T_2 min (s)	Level 2 T_2 min (s)	Level 3 T_2 min (s)
Cat A, C	12	8	5
Cat B	20	8	5

Table 1.4: Minimum time to double amplitude for spiral mode

When the spiral mode is stable ($\lambda_Spiral < 0$), the requirement is automatically satisfied since the amplitude decreases rather than doubles.

1.6.3 Dutch Roll Mode Requirements

The Dutch roll is an oscillatory mode combining yaw and roll motions. Adequate damping of this mode is essential for pilot comfort and precision in lateral control. Three parameters are evaluated: the damping ratio ξ_DR, the product $\xi_DR \cdot \omega_n_DR$, and the natural frequency ω_n_DR.

Flight Phase Category	Level 1 ξ_DR min	Level 1 $\xi \cdot \omega$_n min	Level 1 ω_n min (rad/s)	Level 2 ξ_DR min	Level 2 $\xi \cdot \omega$_n min	Level 2 ω_n min (rad/s)	Level 3 ω_n min (rad/s)
Cat A (Class I, IV)	0.19	0.35	1.0	0.02	0.05	0.5	0.4
Cat A (Class II, III)	0.19	0.35	0.5	0.02	0.05	0.5	0.4
Cat B (all classes)	0.08	0.15	0.5	0.02	0.05	0.5	0.4
Cat C (Class I, IV)	0.08	0.15	1.0	0.02	0.05	0.5	0.4
Cat C (Class II, III)	0.08	0.10	0.5	0.02	0.05	0.5	0.4

Table 1.5: Dutch roll mode flying quality requirements per MIL-F-8785C

For Level 3, only a minimum natural frequency is required ($\omega_{n_{DR}} \geq 0.4$ rad/s for all flight phases), with no minimum damping ratio specified.

1.7 Summary

The MIL-F-8785C specification provides a comprehensive framework for evaluating aircraft flying qualities through the classification of aircraft type, flight phase, and quality level. For longitudinal motion, the short period mode is evaluated using three criteria (damping ratio limits, frequency-damping diagram, and the CAP parameter), while the phugoid mode is evaluated primarily on its damping ratio. For lateral-directional motion, specific requirements are defined for the roll mode time constant, the spiral mode time to double amplitude, and the Dutch roll frequency and damping characteristics.

In the subsequent chapters, these requirements will be applied to assess the flying qualities of the C2 aircraft, after establishing the dynamic stability equations and computing the stability mode characteristics.

Chapter 2: Dynamic Stability of the Aircraft

Understanding the dynamic behavior of an aircraft requires translating the physical laws of mechanics into a mathematical framework adapted to flight. This chapter develops this framework progressively: from the choice of reference frames and the general equations of motion, through the linearization process that makes the problem tractable, to the extraction of the stability modes that characterize how the aircraft responds to perturbations.

2.1 Reference Frames and Motion Variables

2.1.1 The Need for Two Reference Frames

Describing aircraft motion involves a fundamental tension: Newton's laws require an inertial (non-accelerating) reference frame, yet the aerodynamic forces naturally express themselves in a frame attached to the aircraft. This leads to the use of two coordinate systems simultaneously.

The Earth-fixed frame (O_E, x_E, y_E, z_E) serves as the inertial reference. Its origin is fixed on the Earth's surface, with x_E pointing North, y_E pointing East, and z_E pointing downward along the gravity vector. For the flight regimes considered in this study (subsonic, non-polar, non-orbital), the flat-Earth approximation is entirely valid and this frame can be treated as Galilean.

The body-fixed frame (O, x_b, y_b, z_b) is rigidly attached to the aircraft with its origin at the center of gravity. The x_b axis lies along the fuselage reference line (pointing forward), y_b points along the right wing (starboard), and z_b completes the right-handed triad by pointing downward. This frame rotates and translates with the aircraft, which means that while aerodynamic forces have simple expressions in this frame, the equations of motion acquire additional terms from the rotation of the reference itself.

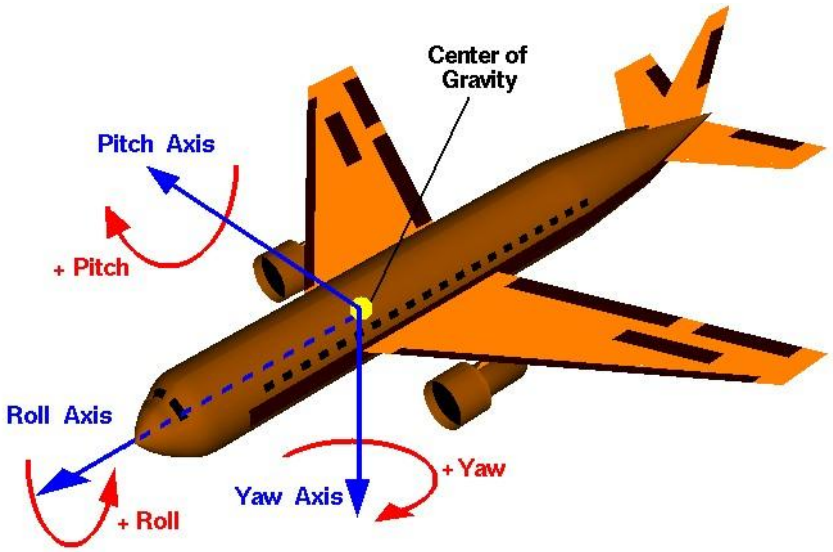


Figure 2.1: Body-fixed axes and rotations of the aircraft. Roll (Φ) about the longitudinal axis, Pitch (Θ) about the lateral axis, and Yaw (Ψ) about the vertical axis. The origin is at the center of gravity.

2.1.2 The Twelve State Variables

A rigid aircraft in free flight possesses six degrees of freedom – three translations and three rotations – requiring twelve state variables for a complete description of its motion. These naturally separate into two groups:

Dynamic variables (governed by Newton's second law):

Variable	Body Axis	Physical Meaning	Units
u, v, w	x_b, y_b, z_b	Linear velocity components of the CG	m/s
p, q, r	x_b, y_b, z_b	Angular velocity components (roll, pitch, yaw rates)	rad/s

Kinematic variables (relating body frame to Earth frame):

Variable	Frame	Physical Meaning	Units
x_E, y_E, z_E	Earth-fixed	Position of the CG	m
Φ, Θ, Ψ	Euler angles	Attitude: bank (roll), pitch, heading (yaw)	rad

The Euler angles deserve particular attention. They describe the orientation of the body frame relative to the Earth frame through a specific sequence of three rotations: first a yaw rotation Ψ about z_E, then a pitch rotation Θ about the intermediate y-axis, and finally a roll rotation Φ about the resulting x-axis. The order of these rotations matters because finite rotations do not commute. The angular velocities in the body frame relate to the Euler angle rates through a transformation that involves trigonometric functions of the attitude angles – a source of nonlinearity in the equations.

The relationship between angular rates in the body frame and Euler angle rates is:

$$p = \dot{\Phi} - \dot{\Psi}\sin\Theta$$

$$q = \dot{\Theta}\cos\Phi + \dot{\Psi}\cos\Theta\sin\Phi$$

$$r = \dot{\Psi}\cos\Theta\cos\Phi - \dot{\Theta}\sin\Phi$$

These kinematic relations, combined with the dynamic equations that follow, form the complete set of twelve first-order differential equations governing the aircraft's motion.

2.2 General Equations of Motion

2.2.1 Force Equations

Newton's second law, F = ma, is straightforward in an inertial frame. However, since we express velocities in the rotating body frame, the time derivative of the velocity vector acquires cross-product terms. If V = (u, v, w) is the velocity of the center of gravity expressed in body axes, and Ω = (p, q, r) is the angular velocity vector, then the inertial acceleration is:

$$a_{inertial} = dV/dt|_{body} + \Omega \times V$$

Expanding the cross product and equating to the applied forces (aerodynamic X, Y, Z and gravitational components) gives the three translational equations:

$$m(\dot{u} + qw - rv) = X - mg\sin\Theta$$

$$m(\dot{v} + ru - pw) = Y + mg\cos\Theta\sin\Phi$$

$$m(\dot{w} + pv - qu) = Z + mg\cos\Theta\cos\Phi$$

The terms qw, rv, ru, pw, pv, qu are the Coriolis-like accelerations arising from the rotation of the body frame. They couple all six degrees of freedom and are a major source of nonlinearity. The gravitational terms on the right-hand side depend on the aircraft attitude through Θ and Φ, introducing further nonlinear coupling between the translational and rotational equations.

2.2.2 Moment Equations

The rotational analogue of Newton's law, applied about the center of gravity in the body frame, yields the moment equations. For a rigid body with a plane of symmetry (the x_b–z_b plane, as is the case for conventional aircraft), the products of inertia I_xy and I_yz vanish, leaving only I_xz as a coupling term between roll and yaw:

$$I_{xx}\dot{p} - I_{xz}\dot{r} + (I_{zz} - I_{yy})qr - I_{xz}pq = L$$

$$I_{yy}\dot{q} + (I_{xx} - I_{zz})pr + I_{xz}(p^2 - r^2) = M$$

$$I_{zz}\dot{r} - I_{xz}\dot{p} + (I_{yy} - I_{xx})pq + I_{xz}qr = N$$

where L, M, N are the aerodynamic and propulsive rolling, pitching, and yawing moments, and I_xx, I_yy, I_zz are the moments of inertia about the body axes.

A crucial observation: the I_xz terms directly couple the roll equation ($\dot{p}$) with the yaw acceleration ($\dot{r}$) and vice versa. For the C2 aircraft, I_xz/I_xx = 0.25, which means this coupling is physically significant – a rolling acceleration will induce a yawing acceleration of about 25% of its magnitude, and conversely. This has practical consequences for the lateral stability analysis that will be addressed in Section 2.5.

2.2.3 The Complete Nonlinear System

Together with the kinematic equations relating body-frame angular rates to Euler angle rates, we obtain a system of twelve coupled, nonlinear, ordinary differential equations. The right-hand side forces and moments X, Y, Z, L, M, N depend on the state variables themselves (through the aerodynamic coefficients), on the control surface deflections (δe, δa, δr, δT), and on atmospheric disturbances. This system has no general analytical solution, which motivates the linearization approach described next.

2.3 Small Perturbation Theory and Linearization

2.3.1 The Perturbation Concept

The key idea is to decompose each variable into a known steady-state (equilibrium) value and a small, unknown perturbation:

$$U = U_0 + \Delta u, \quad V = V_0 + \Delta v, \quad W = W_0 + \Delta w$$

$$P = P_0 + \Delta p, \quad Q = Q_0 + \Delta q, \quad R = R_0 + \Delta r$$

The forces and moments are likewise expanded using a first-order Taylor series about the equilibrium. For instance, the aerodynamic force X, which depends on u, w, ẇ, q, and the controls, is expanded as:

$$\Delta X = (\partial X/\partial u)\Delta u + (\partial X/\partial w)\Delta w + (\partial X/\partial \dot{w})\Delta \dot{w} + (\partial X/\partial q)\Delta q + (\partial X/\partial \delta e)\Delta \delta e + \ldots$$

The partial derivatives $\partial X/\partial u$, $\partial X/\partial w$, etc., evaluated at the equilibrium condition, are the stability and control derivatives. They quantify how sensitive each force or moment component is to a change in each state variable. These derivatives are the fundamental building blocks of the linear model.

2.3.2 Equilibrium Conditions

For steady, rectilinear, wings-level flight the equilibrium conditions are:

- $V_0 = 0$: no lateral velocity (symmetric flight)
- $P_0 = Q_0 = R_0 = 0$: no angular rates (steady, non-rotating)
- $\Phi_0 = 0$: wings level, therefore $\sin\Phi_0 = 0$ and $\cos\Phi_0 = 1$
- Θ_0 = constant : steady pitch angle
- For the C2 aircraft: $\Theta_0 = 0°$, which further simplifies the equations since $\cos\Theta_0 = 1$ and $\sin\Theta_0 = 0$.

With $W_0 = U_0 \tan(\alpha_0) \approx 0$ for the C2 (since $\alpha_0 = 0°$), the body x-axis is aligned with the flight direction.

2.3.3 The Linearization Process

Substituting the perturbed variables into the nonlinear equations and applying the following rules produces the linearized system:

- Products of two small perturbations are negligible: $\Delta q \cdot \Delta w \approx 0$, $\Delta p \cdot \Delta r \approx 0$, etc.
- Trigonometric functions of small perturbation angles are linearized: $\sin(\Theta_0 + \Delta\theta) \approx \sin\Theta_0 + \Delta\theta \cos\Theta_0$, $\cos(\Theta_0 + \Delta\theta) \approx \cos\Theta_0 - \Delta\theta \sin\Theta_0$.
- The steady-state equation (equilibrium forces balance gravity) is subtracted out.

A critical result of this linearization, which relies on the aircraft's plane of symmetry, is the decoupling of the equations into two independent sets:

- Longitudinal dynamics: perturbations Δu, Δw, Δq, $\Delta\theta$ (symmetric motion in the plane of symmetry, driven by elevator δe and thrust δT).
- Lateral-directional dynamics: perturbations Δv, Δp, Δr, $\Delta\varphi$ (asymmetric motion involving sideslip and roll/yaw, driven by ailerons δa and rudder δr).

This decoupling is physically meaningful: in the absence of asymmetric disturbances, a symmetric aircraft disturbed in the longitudinal plane will not develop lateral motion, and vice versa. It reduces the original 12th-order nonlinear system into two manageable 4th-order linear systems.

2.4 Longitudinal Equations of Motion

2.4.1 The Four Linearized Equations

After linearization, the longitudinal motion is described by four equations. The first three come from the x-force, z-force, and pitching moment equations respectively, while the fourth is the kinematic relation between pitch rate and pitch angle:

Axial force equation (along x_b):

$$(d/dt - X_u)\Delta u - X_w\Delta w + g\cos\Theta_0\,\Delta\theta = X_{\delta e}\,\Delta\delta e + X_{\delta T}\,\Delta\delta T$$

This equation states that the acceleration of the aircraft along its longitudinal axis depends on the drag perturbation ($X_u\Delta u$, $X_w\Delta w$) and on the component of gravity that appears when the aircraft pitches ($\Delta\theta$). The terms $X_u < 0$ provides natural speed stability: if the aircraft accelerates, the increased drag decelerates it.

Normal force equation (along z_b):

$$-Z_u\Delta u + (d/dt - Z_w - Z_{\dot w}\,d/dt)\Delta w - (Z_q + U_0)\Delta q + g\sin\Theta_0\,\Delta\theta = Z_{\delta e}\,\Delta\delta e$$

The key term here is $(Z_q + U_0)\Delta q$, where U_0 dominates: a pitch rate Δq at speed U_0 curves the flight path and changes the effective angle of attack, generating a large normal force. The derivative $Z_{\dot w}$ accounts for the lag in downwash reaching the horizontal tail when the wing angle of attack changes.

Pitching moment equation:

$$-M_u\Delta u - (M_w + M_{\dot{w}}\, d/dt)\Delta w + (d/dt - M_q)\Delta q = M_{\delta e}\,\Delta\delta e$$

The derivative M_w (or equivalently C_mα when non-dimensionalized) is the static stability derivative: M_w < 0 means that an increase in angle of attack generates a nose-down restoring moment. The derivative M_q < 0 provides pitch damping: a nose-up pitch rate generates a nose-down moment, opposing the rotation.

Kinematic relation:

$$d(\Delta\theta)/dt = \Delta q$$

2.4.2 State-Space Formulation

Collecting these equations into matrix form gives the state-space representation:

$$\dot{X}_{Long} = A_{Long} \cdot X_{Long} + B_{Long} \cdot \eta_{Long}$$

with state and control vectors:

$$X_{Long} = [\Delta u, \Delta w, \Delta q, \Delta\theta]^T$$

$$\eta_{Long} = [\Delta\delta e, \Delta\delta T]^T$$

The raw equations contain a mass-like matrix I_n on the left-hand side (due to the Z_ẇ and M_ẇ terms that involve dΔw/dt). Inverting this matrix and applying the simplifications Z_ẇ ≪ 1 and Z_q ≪ U_0 yields the standard A and B matrices:

A_Long =

X_u	X_w	0	$-g\cos\Theta_0$
Z_u	Z_w	U_0	$-g\sin\Theta_0$
M_u + Z_u·M_ẇ	M_w + Z_w·M_ẇ	M_q + U_0·M_ẇ	0
0	0	1	0

B_Long =

X_δe	X_δT
Z_δe	Z_δT
M_δe + Z_δe·M_ẇ	M_δT + Z_δT·M_ẇ
0	0

The third row of A_Long illustrates the effect of the I_n matrix inversion: the M_ẇ derivative creates additional coupling terms (Z_u·M_ẇ, Z_w·M_ẇ, U_0·M_ẇ) that would not appear if the downwash lag were neglected.

2.4.3 Dimensional Stability Derivatives

The dimensional derivatives appearing in A_Long and B_Long are computed from the non-dimensional aerodynamic coefficients. The conversion involves the dynamic pressure $\bar{q} = \frac{1}{2}\rho U_0^2$, the aircraft mass m, and the relevant inertia and geometric parameters. The formulas reflect the physical origin of each derivative:

Derivatives along the x-axis (drag-related):

$$X_u = -(\bar{q}\, S)/(m\, U_0) \cdot (2C_{D_0} + C_{Du})$$

The factor $2C_{D_0}$ arises because drag varies as U^2, so a speed perturbation changes drag both through the dynamic pressure (proportional to $2U_0\Delta u$) and through any explicit Mach dependence (C_{Du}).

$$X_w = (\bar{q}\, S)/(m\, U_0) \cdot C_{L_0}(1 - 2C_{L\alpha}/(\pi e A))$$

This derivative reflects the induced drag change: when the angle of attack increases, the lift increases, which increases the induced drag. The term $2C_{L\alpha}/(\pi e A)$ is the induced drag contribution, scaled by the Oswald efficiency and aspect ratio.

Derivatives along the z-axis (lift-related):

$$Z_u = -(\bar{q}\, S)/(m\, U_0) \cdot (2C_{L_0} + C_{Lu})$$

$$Z_w = -(\bar{q}\, S)/(m\, U_0) \cdot (C_{D_0} + C_{L\alpha})$$

Z_w is typically the largest longitudinal derivative (in magnitude). For the C2 aircraft, $C_{L\alpha} = 5.5$ dominates over $C_{D_0} = 0.022$, so Z_w is essentially proportional to the lift curve slope.

$$Z_{\dot{w}} = (\bar{q}\, S\, \bar{c})/(2m\, U_0^2) \cdot (C_{D_0} + C_{L\alpha})$$

$$Z_q = (\bar{q}\, S\, \bar{c})/(2m\, U_0) \cdot C_{Lq}$$

Pitching moment derivatives:

$$M_u = (\bar{q}\, S\, \bar{c})/(I_{yy}\, U_0) \cdot C_{mu}$$

$$M_w = (\bar{q}\, S\, \bar{c})/(I_{yy}\, U_0) \cdot C_{m\alpha}$$

$$M_{\dot{w}} = (\bar{q}\, S\, \bar{c}^2)/(2\, I_{yy}\, U_0^2) \cdot C_{m\dot{\alpha}}$$

$$M_q = (\bar{q}\, S\, \bar{c}^2)/(2\, I_{yy}\, U_0) \cdot C_{mq}$$

Elevator control derivatives:

$$X_{\delta e} = (\bar{q}\, S)/(m\, U_0) \cdot C_{D\delta e}$$

$$Z_{\delta e} = -(\bar{q}\, S)/(m\, U_0) \cdot C_{L\delta e}$$

$$M_{\delta e} = (\bar{q}\, S\, \bar{c})/(I_{yy}\, U_0) \cdot C_{m\delta e}$$

Note the pattern: force derivatives scale with $\bar{q}S/(mU_0)$, while moment derivatives scale with $\bar{q}S\bar{c}/(I_{yy}U_0)$. Rate derivatives (involving q or α) gain an additional $\bar{c}/(2U_0)$ factor, reflecting the time a perturbation takes to travel one chord length.

2.5 Longitudinal Stability Modes

2.5.1 Free Response and Characteristic Equation

The stability of the aircraft is analyzed through its free response – the behavior following a perturbation with no control input (η = 0):

$$\dot{X}_{Long} = A_{Long} \cdot X_{Long}$$

Assuming solutions of the form X(t) = V·e^(λt), where V is a constant vector and λ is a scalar, substitution yields the eigenvalue problem:

$$(\lambda I - A_{Long})\, V = 0$$

For non-trivial solutions, the determinant must vanish, giving the characteristic equation:

$$det(\lambda I - A_{Long}) = \lambda^4 + a\lambda^3 + b\lambda^2 + c\lambda + d = 0$$

This 4th-degree polynomial has four roots (eigenvalues) λ_i. For a conventional, statically stable aircraft, these roots form two complex conjugate pairs, each representing a damped oscillatory mode:

$$(\lambda^2 + 2\xi_{SP}\omega_{nSP}\lambda + \omega^2_{nSP})(\lambda^2 + 2\xi_{Ph}\omega_{nPh}\lambda + \omega^2_{nPh}) = 0$$

Each complex eigenvalue pair takes the form:

$$\lambda = \omega_n(-\xi \pm j\sqrt{(1 - \xi^2)})$$

where ω_n is the natural frequency (rad/s), ξ is the damping ratio (dimensionless), and the eigenvector V associated with each eigenvalue describes which state variables participate in that mode and with what relative amplitude and phase.

The mode characteristics are extracted from the eigenvalue pairs as:

$$\omega_n = |\lambda| = \sqrt{(Re^2 + Im^2)}$$

$$\xi = -Re(\lambda) / \omega_n$$

$$Period = 2\pi / (\omega_n\sqrt{(1 - \xi^2)})$$

$$Time\ to\ half\ amplitude = ln(2) / |\xi\omega_n|$$

2.5.2 Short Period Mode

The short period mode is the fast longitudinal oscillation, with a period typically of a few seconds. Physically, it represents the aircraft's transient response in angle of attack following an elevator deflection. The mechanism is straightforward: a change in angle of attack alters the pitching moment (through C_mα), which rotates the aircraft, which changes the angle of attack

further. The pitch damping derivative C_mq provides the restoring force that prevents unbounded oscillation.

The eigenvectors confirm that this mode primarily involves Δw (vertical velocity, proportional to angle of attack) and Δq (pitch rate), with negligible participation of Δu (forward speed remains essentially constant over the short period timescale).

Short period approximation:

Setting Δu = 0 (constant speed) reduces the system to 2×2, involving only Δw and Δq:

A_SP =

Z_w	U_0
$M_w + Z_w \cdot M_{\dot{w}}$	$M_q + U_0 \cdot M_{\dot{w}}$

The closed-form expressions for the natural frequency and damping ratio are:

$$\omega^2_{nSP} = Z_w \cdot M_q - U_0 \cdot M_w$$

Since $Z_w < 0$, $M_q < 0$, and $M_w < 0$ (for a statically stable aircraft), both terms contribute positively, giving a real positive ω^2_{nSP}. The dominant term is usually $-U_0 M_w$, meaning the natural frequency is primarily governed by the pitch stiffness $C_{m\alpha}$ and the flight speed.

$$2\xi_{SP}\omega_{nSP} = -(M_q + Z_w + M_{\dot{w}} \cdot U_0)$$

The damping comes from three sources: pitch rate damping M_q (dominant), the aerodynamic damping Z_w (angle of attack changes that are self-limiting), and the downwash lag effect $M_{\dot{w}} \cdot U_0$. At high altitudes, the aerodynamic damping diminishes (as air density decreases), potentially requiring a Stability Augmentation System.

2.5.3 Phugoid Mode

The phugoid is the slow longitudinal oscillation, with a period of the order of a minute. It involves an energy exchange between kinetic and potential energy at nearly constant angle of attack: the aircraft alternately climbs (losing speed, gaining altitude) and descends (gaining speed, losing altitude). This is fundamentally a trajectory oscillation rather than an attitude oscillation.

Phugoid approximation:

Setting Δw = 0 (constant angle of attack) and neglecting pitch rate dynamics, the system reduces to 2×2 involving Δu and Δθ:

A_Ph =

X_u	$-g \cos\Theta_0$
$-Z_u / U_0$	$g \sin\Theta_0 / U_0$

$$\omega^2_{nPh} = -(Z_u / U_0) \cdot g \cos\Theta_0 - X_u \cdot g \sin\Theta_0 / U_0$$

$$2\xi_{Ph}\omega_{nPh} = -(X_u + g \sin\Theta_0 / U_0)$$

For level flight ($\Theta_0 = 0$), these simplify considerably:

$$\omega^2_{nPh} = -g\, Z_u / U_0 \qquad ; \qquad 2\xi_{Ph}\omega_{nPh} = -X_u$$

The phugoid frequency depends on Z_u (how lift changes with speed) and is roughly proportional to g/U₀ – faster aircraft have slower phugoids. The damping depends entirely on X_u (the speed stability), which itself is governed by the drag coefficient. Since drag is relatively small compared to lift, phugoid damping is inherently low.

2.6 Lateral-Directional Equations of Motion

2.6.1 Linearized Lateral Equations

The lateral motion involves sideslip (Δv), roll (Δp), yaw (Δr), and bank angle (Δφ). After linearization, four equations govern this motion:

$$(d/dt - Y_v)\Delta v - Y_p\Delta p - (Y_r - U_0)\Delta r - g \cos\Theta_0\, \Delta\varphi = Y_{\delta r}\Delta\delta r + Y_{\delta a}\Delta\delta a$$

$$-L_v\Delta v + (d/dt - L_p)\Delta p - L_r\Delta r = L_{\delta r}\Delta\delta r + L_{\delta a}\Delta\delta a$$

$$-N_v\Delta v - N_p\Delta p + (d/dt - N_r)\Delta r = N_{\delta r}\Delta\delta r + N_{\delta a}\Delta\delta a$$

$$d(\Delta\varphi)/dt = \Delta p$$

2.6.2 Inertial Coupling Correction

The roll and yaw moment equations contain an I_xz coupling term that couples the roll and yaw accelerations. To obtain the state-space form, the system must be pre-multiplied by the inverse of an inertial coupling matrix I_n:

I_n =

1	0	0	0
0	1	−I_xz/I_xx	0
0	−I_xz/I_zz	1	0
0	0	0	1

The state-space form is then obtained from:

$$\dot{X}_{Lat} = I_n^{-1} \cdot A_{n_Lat} \cdot X_{Lat} + I_n^{-1} \cdot B_{n_Lat} \cdot \eta_{Lat}$$

where A_n_Lat and B_n_Lat are the matrices obtained before inversion (with the simple derivatives $L_v = (\bar{q}Sb)/(I_{xx} U_0)\cdot C_{l\beta}$, $N_v = (\bar{q}Sb)/(I_{zz} U_0)\cdot C_{n\beta}$, etc.). The multiplication by I_n^{-1} naturally introduces the I_xz cross-coupling into all rolling and yawing derivatives.

When I_xz is negligible ($I_{xz} \ll I_{xx}$ and $I_{xz} \ll I_{zz}$), the matrix I_n reduces to the identity and the coupling vanishes. For the C2 aircraft however, $I_{xz}/I_{xx} = 0.25$ is significant, and this inversion is retained throughout the analysis to properly capture the roll–yaw inertial coupling.

2.6.3 State-Space Formulation

The lateral state-space form is:

$$\dot{X}_Lat = A_Lat \cdot X_Lat + B_Lat \cdot \eta_Lat$$

with:

$$X_Lat = [\Delta v, \Delta p, \Delta r, \Delta\varphi]^T \quad ; \quad \eta_Lat = [\Delta\delta r, \Delta\delta a]^T$$

A_n_Lat =

Y_v	Y_p	$Y_r - U_0$	$g\cos\Theta_0$
L_v	L_p	L_r	0
N_v	N_p	N_r	0
0	1	0	0

B_n_Lat =

Y_δr	Y_δa
L_δr	L_δa
N_δr	N_δa
0	0

The final matrices A_Lat and B_Lat used in the state-space equation are obtained by pre-multiplication with I_n^{-1}:

$$A_Lat = I_n^{-1} \cdot A_n_Lat \quad ; \quad B_Lat = I_n^{-1} \cdot B_n_Lat$$

The Y-force derivatives (first row of A_n_Lat and B_n_Lat) are unaffected by the inversion since they come from the force equation. Only the rolling and yawing moment derivatives (rows 2 and 3) are modified by the I_xz coupling.

2.6.4 Lateral Dimensional Derivatives

Side force derivatives:

$$Y_v = (\bar{q}\, S)/(m\, U_0) \cdot C_Y\beta$$

$$Y_p = (\bar{q}\, S\, b)/(2m\, U_0) \cdot C_{Yp} \quad ; \quad Y_r = (\bar{q}\, S\, b)/(2m\, U_0) \cdot C_{Yr}$$

Rolling moment derivatives:

$$L_v = (\bar{q}\, S\, b)/(I_{xx}\, U_0) \cdot C_{l\beta}$$

$$L_p = (\bar{q}\, S\, b^2)/(2\, I_{xx}\, U_0) \cdot C_{lp} \quad ; \quad L_r = (\bar{q}\, S\, b^2)/(2\, I_{xx}\, U_0) \cdot C_{lr}$$

Yawing moment derivatives:

$$N_v = (\bar{q}\, S\, b)/(I_{zz}\, U_0) \cdot C_{n\beta}$$

$$N_p = (\bar{q}\, S\, b^2)/(2\, I_{zz}\, U_0) \cdot C_{np} \quad ; \quad N_r = (\bar{q}\, S\, b^2)/(2\, I_{zz}\, U_0) \cdot C_{nr}$$

These derivatives populate the A_n_Lat matrix shown above. The actual state-space matrix A_Lat is then obtained by multiplying by I_n^{-1} to account for the I_xz inertial coupling.

2.7 Lateral Stability Modes

The lateral characteristic equation $\det(\lambda I - A_{Lat}) = 0$ yields a 4th-degree polynomial that, unlike the longitudinal case, typically factors into two real roots and one complex conjugate pair:

$$(\lambda - \lambda_{Roll})(\lambda - \lambda_{Spiral})(\lambda^2 + 2\xi_{DR}\omega_{nDR}\lambda + \omega^2_{nDR}) = 0$$

2.7.1 Roll Subsidence Mode

The roll mode is a non-oscillatory, heavily damped exponential decay in roll rate. When the pilot deflects the ailerons, the aircraft builds up roll rate; when the ailerons return to neutral, the roll damping derivative C_lp brings the roll rate to zero within a fraction of a second. The mode barely involves sideslip or yaw.

Approximation (pure roll, $\Delta v = \Delta r = 0$):

$$\lambda_{Roll} = L'_p \quad ; \quad T_{Roll} = -1/\lambda_{Roll}$$

L'_p is always large and negative, giving a very short time constant. For the C2, the approximation $L'_p \approx -7.47$ gives $T_{Roll} \approx 0.13$ s.

2.7.2 Spiral Mode

The spiral mode is the slowest lateral mode – an exponential that can be either convergent or divergent. Physically, it manifests as a gradual, uncommanded bank angle change. When divergent, the aircraft slowly rolls into a banked turn; when convergent, it returns to wings-level. The stability of this mode depends on the balance between two competing effects: dihedral stability (C_lβ, which opposes sideslip-induced roll) and directional stability (C_nβ, which weathervanes the nose into the relative wind). The classical stability criterion is $C_{l\beta} \cdot C_{nr} - C_{n\beta} \cdot C_{lr}$.

Approximation ($d\Delta p/dt \approx 0$):

$$\lambda_{Spiral} = N'_r - N'_v \cdot (L'_r / L'_v)$$

The spiral mode is often slightly unstable, and this is tolerated provided the time to double amplitude is long enough for the pilot to correct. The pilot barely notices a spiral divergence with $T_2 > 20$ seconds.

2.7.3 Dutch Roll Mode

The Dutch roll is the oscillatory lateral mode, combining yawing and rolling motions in a characteristic pattern where the aircraft's nose traces an elliptical path. The name comes from its resemblance to the motion of a speed skater. The primary physical driver is directional stability (C_nβ): when the aircraft yaws, the fin generates a restoring moment that swings the nose back, overshoots, and oscillates. Roll participates because the yawing motion changes the angle of attack of each wing differently, generating a rolling moment.

Approximation (rolling suppressed, Δp = 0):

A_DR =

Y_v	−(U₀ − Y_r)
N'_v	N'_r

$$\omega^2_{nDR} = Y_v \cdot N'_r + N'_v \cdot (U_0 - Y_r)$$

$$2\xi_{DR}\omega_{nDR} = -(Y_v + N'_r)$$

The Dutch roll frequency is dominated by the term $N'_v \cdot U_0$, which is proportional to $C_{n\beta} \cdot U_0$: greater directional stability or higher speed means a faster Dutch roll. The damping comes from side force damping Y_v (which dissipates sideslip energy) and yaw damping N'_r (which opposes yaw rate). Both are typically small, which is why Dutch roll damping is often the most critical lateral flying quality parameter.

2.8 Summary

Mode	Type	Primary Variables	Physical Mechanism	Typical Period	Damping
Short Period	Oscillatory	Δw, Δq	Pitch stiffness (C_mα) vs pitch damping (C_mq)	~ 1 s	High
Phugoid	Oscillatory	Δu, Δθ	Kinetic ↔ potential energy exchange	~ 1 min	Low
Roll	Exponential	Δp	Roll damping (C_lp)	< 1 s	Very high
Spiral	Exponential	Δφ, Δr	Dihedral vs directional	~ mins	Very low

			stability balance		
Dutch Roll	Oscillatory	Δv, Δr, Δp	Directional stability ($C_{n\beta}$) oscillation	~ secs	Moderate

Table 2.1: Summary of the five aircraft stability modes

The next chapter applies these formulations numerically to the C2 aircraft, computing all matrices, eigenvalues, and mode characteristics, and comparing them to the flying quality requirements established in Chapter 1.

Chapter 3: Application to the C2 Aircraft

This chapter applies the theoretical framework developed in Chapter 2 to the C2 aircraft using the geometric, inertial and aerodynamic data provided. All computations are performed in MATLAB (code in Appendix A). The stability mode characteristics are obtained through the eigenvector decomposition method, the transfer functions are computed from the state-space matrices, and the aircraft is assessed against the MIL-F-8785C flying quality requirements.

3.1 C2 Aircraft Data

3.1.1 Geometry and Flight Condition

Parameter	Value	Parameter	Value
S (wing area)	12.63 m²	Altitude	7620 m
$\bar{c}$ (MAC)	1.65 m	Mach	0.6
b (wingspan)	8.02 m	U_0 (airspeed)	185.9 m/s
A (aspect ratio)	5.09	$\bar{q}$ (dynamic pressure)	9500 Pa
e (Oswald)	0.94	ρ (air density)	0.55 kg/m³
C_L0 / C_D0	0.15 / 0.022	Θ_0	0°

3.1.2 Mass and Inertia

m (kg)	I_xx	I_yy	I_zz	I_xz	I_xz / I_xx
1816	1085	6511	7053	271	0.25

Note: The ratio I_xz/I_xx = 0.25 is not negligible, so the inertial coupling between roll and yaw motion is retained through the inertial matrix I_n inversion described in Chapter 2, Section 2.6.2.

3.2 Aircraft and Control Matrices

3.2.1 Longitudinal Dimensional Derivatives

The longitudinal dimensional stability derivatives are obtained from the non-dimensional aerodynamic coefficients using the formulas of Chapter 2:

X_u	X_w	Z_u	Z_w	Z_ẇ
−0.0156	0.0143	−0.1365	−1.9626	0.0067
M_u	**M_w**	**M_ẇ**	**M_q**	**M_δe**
0.0000	−0.0393	−0.0070	−2.3884	−0.1439

3.2.2 Longitudinal State-Space Matrices

Inserting the numerical values into the state-space form:

$$\dot{X}_Long = A_Long \cdot X_Long + B_Long \cdot \eta_Long$$

A_Long =

−0.0156	0.0143	0	−9.8100
−0.1365	−1.9626	185.9000	0
0.0010	−0.0256	−3.6838	0
0	0	1.0000	0

B_Long =

0
−0.0135
−0.1438
0

Note: only the elevator column is shown, as no thrust control derivatives were provided for the C2 aircraft.

3.2.3 Lateral Dimensional Derivatives and I_n Inversion

The lateral dimensional derivatives are first computed using the simple formulas (divided by I_xx or I_zz), which constitute the matrix A_n_Lat before I_xz coupling:

Derivative	Value	Derivative	Value
Y_v	−0.3554	L_δr	0.2385
L_v	−0.5248	N_δr	−0.0881
N_v	0.1248	L_δa	0.4771

The final state-space matrix A_Lat is then obtained by pre-multiplication with the inverse of the inertial matrix I_n:

$$A_Lat = I_n^{-1} \cdot A_n_Lat \qquad ; \qquad B_Lat = I_n^{-1} \cdot B_n_Lat$$

I_n =

1	0	0	0
0	1	−0.2498	0
0	−0.0384	1	0

0	0	0	1

3.2.4 Lateral State-Space Matrices

After I_n inversion, the lateral state-space matrices become:

A_Lat =

−0.3554	−0.1995	−185.0306	9.8100
−0.4984	−7.4666	5.2156	0
0.1056	−0.0220	−0.5648	0
0	1.0000	0	0

B_Lat =

0.0995	0
0.2186	0.4811
−0.0797	0.0163
0	0

The I_n inversion has a noticeable effect on the N-row derivatives. In particular, N_p changes sign (from +0.265 before inversion to −0.022 after), reflecting the strong roll–yaw inertial coupling in the C2 aircraft.

3.3 Characteristic Equations and Eigenvalues

3.3.1 Longitudinal Characteristic Equation

Computing det(λI − A_Long) = 0 yields the 4th-degree polynomial:

$$\lambda^4 + 5.6621\lambda^3 + 12.0752\lambda^2 + 0.2014\lambda + 0.0526 = 0$$

The four eigenvalues form two complex conjugate pairs:

Mode	Eigenvalues	Associated variables
Short Period	$\lambda_{1,2} = -2.8237 \pm 2.0036j$	Δw, Δq
Phugoid	$\lambda_{3,4} = -0.0074 \pm 0.0658j$	Δu, Δθ

3.3.2 Lateral Characteristic Equation

Computing det(λI − A_Lat) = 0 yields:

$$\lambda^4 + 8.3868\lambda^3 + 26.6290\lambda^2 + 154.4261\lambda - 2.6424 = 0$$

The negative constant term (−2.6424) indicates the presence of a positive real root, which corresponds to an unstable spiral mode. The factored form gives:

Mode	Eigenvalue(s)	Associated variables
Dutch Roll	$\lambda_{1,2} = -0.4169 \pm 4.5040j$	Δv, Δp, Δr
Roll	$\lambda_3 = -7.5700$ (real)	Δp
Spiral	$\lambda_4 = +0.0171$ (real, unstable)	Δφ, Δr

3.4 Stability Mode Characteristics

3.4.1 Longitudinal Modes

Short Period Mode:

- ω_n_SP = 3.4623 rad/s
- ξ_SP = 0.8155
- Period = 3.14 s, Time to half amplitude = 0.25 s

Phugoid Mode:

- ω_n_Ph = 0.0662 rad/s
- ξ_Ph = 0.1113
- Period = 95.5 s, Time to half amplitude = 94 s

3.4.2 Lateral Modes

Dutch Roll Mode:

- ω_n_DR = 4.5233 rad/s
- ξ_DR = 0.0922
- ξ_DR · ω_n_DR = 0.4169
- Period = 1.40 s

Roll Mode:

- λ_Roll = −7.5700
- T_Roll = 0.132 s

Spiral Mode:

- λ_Spiral = +0.0171 (unstable)
- T_2 (time to double amplitude) = 40.6 s

3.4.3 Comparison with Approximations

The results obtained from the full 4×4 matrices are compared with the 2×2 approximations derived in Chapter 2:

Mode	Exact ω_n	Approx ω_n	Exact ξ	Approx ξ
Short Period	3.4623	3.4619	0.8155	0.8155
Phugoid	0.0662	0.0849	0.1113	0.0921
Roll (λ)	−7.5700	−7.4666	—	—
Spiral (λ)	+0.0171	+0.5404	—	—
Dutch Roll	4.5233	4.4433	0.0922	0.1035

The Short Period, Roll, and Dutch Roll approximations match the exact values within a few percent. The Phugoid approximation overestimates the natural frequency by ~28%, a known limitation of the constant-angle-of-attack assumption. The Spiral approximation deviates more significantly from the exact eigenvalue, since this mode depends on a delicate balance between the lateral derivatives.

3.5 State Variable Time Responses

The time response of each stability mode is plotted using the eigenvector decomposition method. For each mode, the four state variables are reconstructed from the corresponding eigenvalue and eigenvector:

$$X(t) = 2 \cdot Re(V_mode \cdot e^{(\lambda_mode \cdot t)})$$

The amplitudes are normalized to unity for visualization, which highlights the dominant variable in each mode.

3.5.1 Short Period Mode

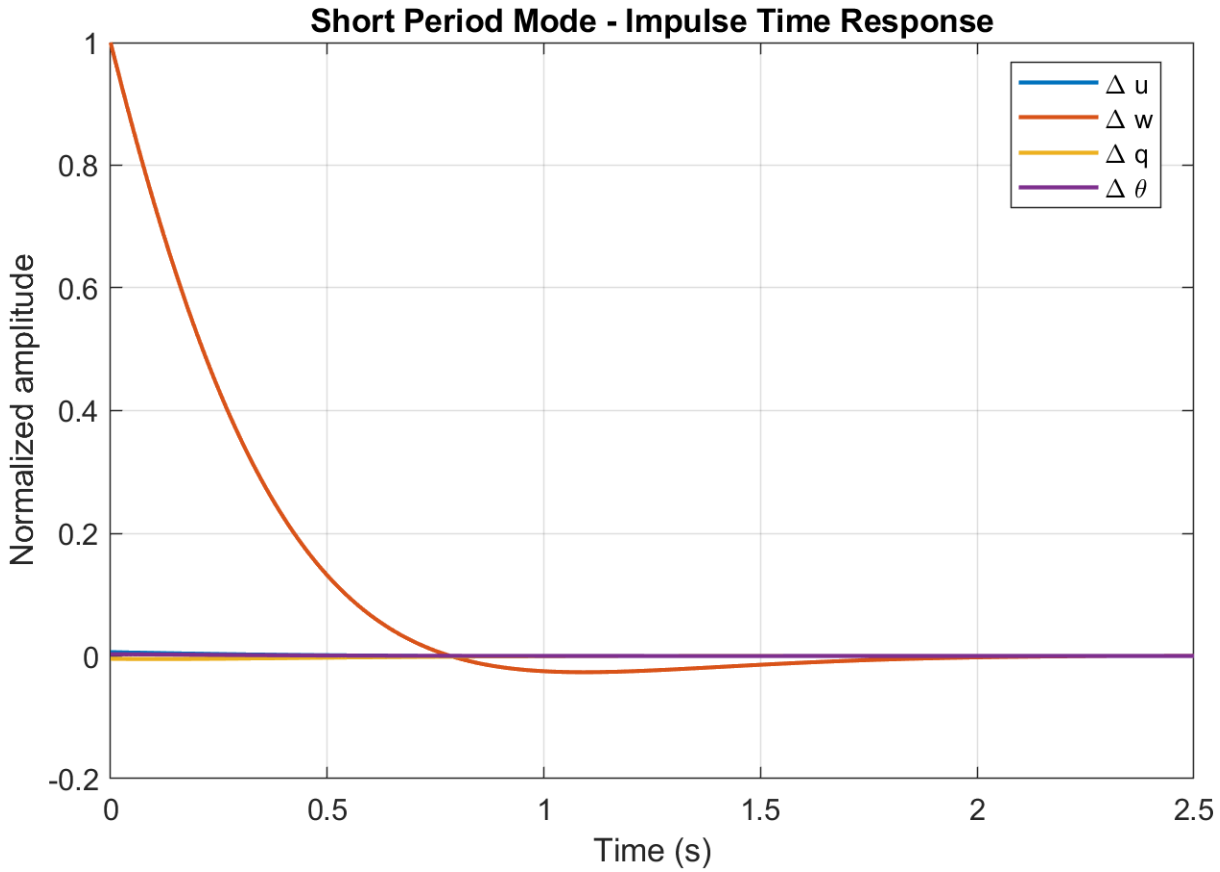


Figure 3.1: Short Period Mode – Impulse Time Response (normalized).

The angle of attack Δw (orange) is completely dominant and decays from 1 to nearly zero within 2 seconds, confirming the near-critical damping (ξ_SP = 0.82). The other variables Δu (axial velocity), Δq (pitch rate) and Δθ (pitch angle) remain negligible on the normalized scale. This is the textbook behavior of the short period mode: a fast incidence transient driven by the static stability derivative C_mα, damped by the pitch rate derivative C_mq.

3.5.2 Phugoid Mode

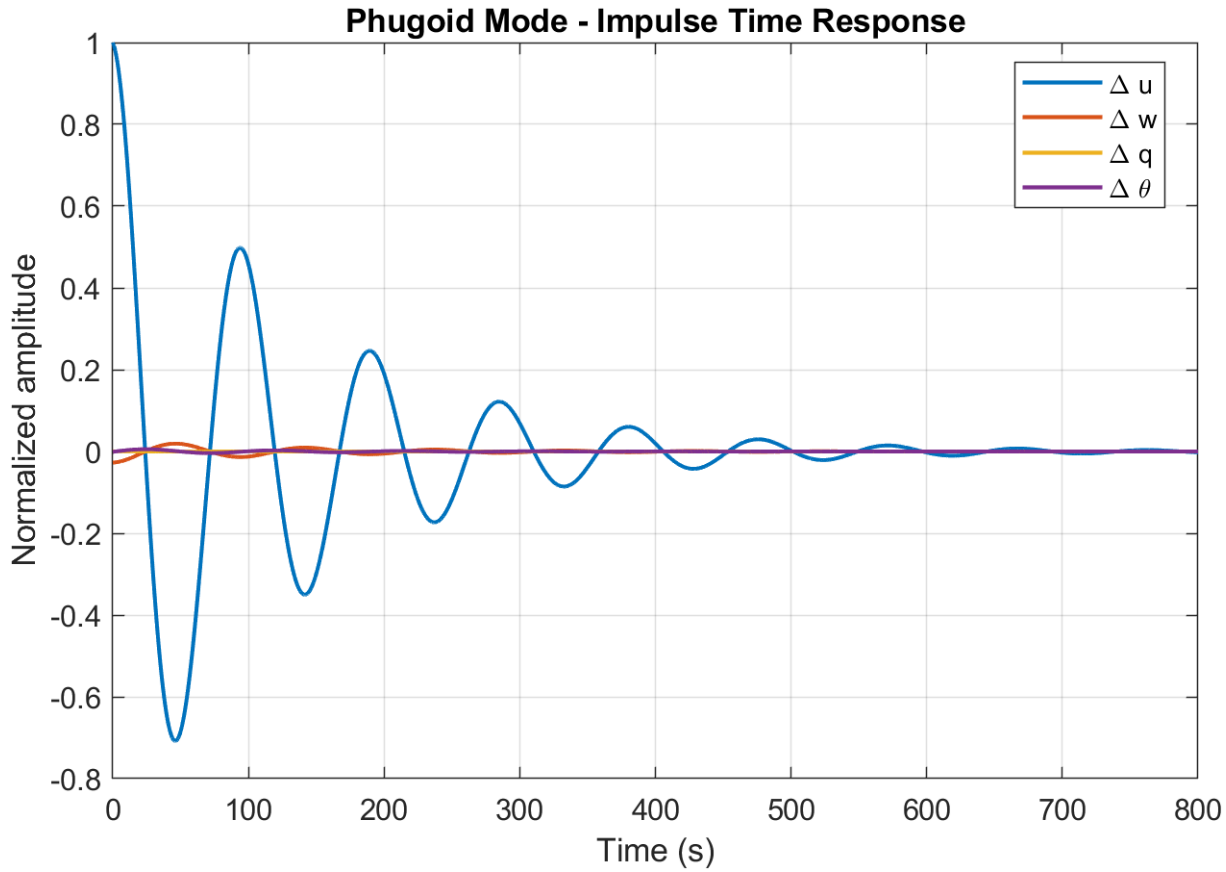


Figure 3.2: Phugoid Mode – Impulse Time Response (normalized).

The axial velocity Δu (blue) is dominant and oscillates over a long period (~95 s) with slow damping (ξ_Ph = 0.11). The pitch angle Δθ shows a barely visible correlated oscillation, while Δw and Δq remain essentially zero. This confirms that the phugoid occurs at nearly constant angle of attack and is a kinetic–potential energy exchange: the aircraft alternately climbs (slowing down) and dives (speeding up) while maintaining its lift.

3.5.3 Dutch Roll Mode

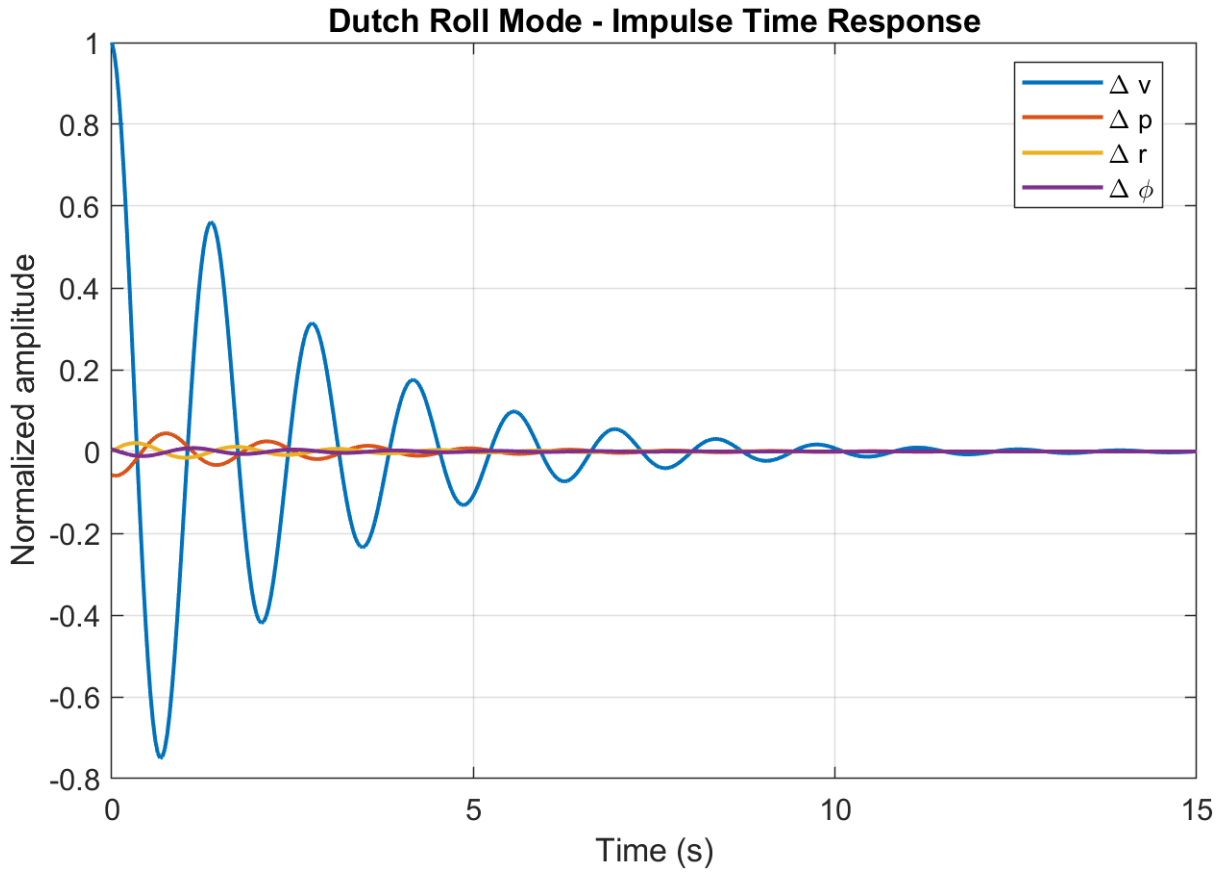


Figure 3.3: Dutch Roll Mode – Impulse Time Response (normalized).

The sideslip velocity Δv (blue) is the dominant variable, oscillating at the Dutch roll frequency (period 1.4 s) and decaying with ξ_DR = 0.09. The roll rate Δp (orange) and yaw rate Δr (yellow) also participate (visible as small coupled oscillations), which illustrates the combined yaw–roll nature of the Dutch roll. The oscillation is essentially damped within 15 seconds.

3.5.4 Roll Mode

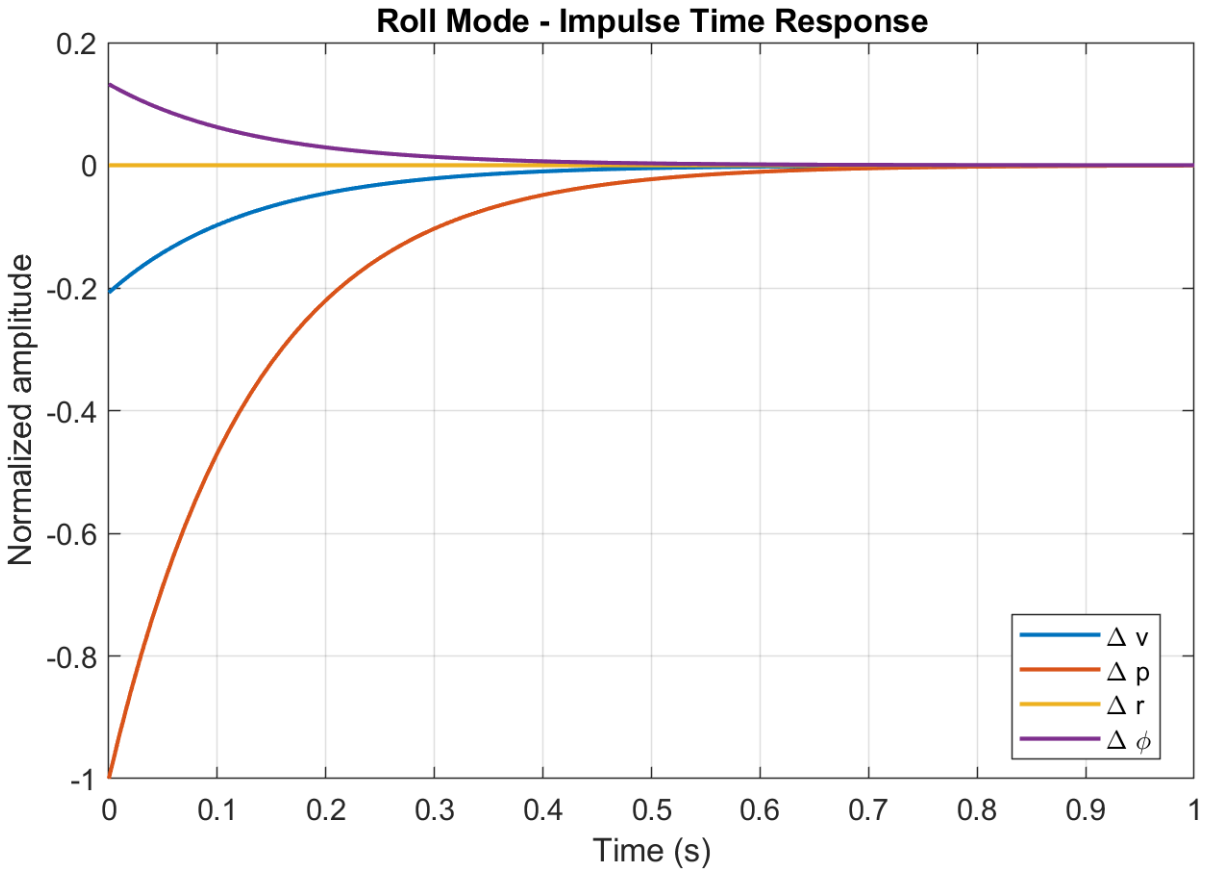


Figure 3.4: Roll Mode – Impulse Time Response (normalized).

The roll rate Δp (orange) is dominant and decays exponentially within 0.5 s, consistent with the time constant T_Roll = 0.13 s. The bank angle Δφ (purple) follows as its integral and relaxes similarly. The sideslip Δv and yaw rate Δr remain near zero, confirming the pure-roll character of this highly damped mode, which is governed by the roll damping derivative C_lp.

3.5.5 Spiral Mode

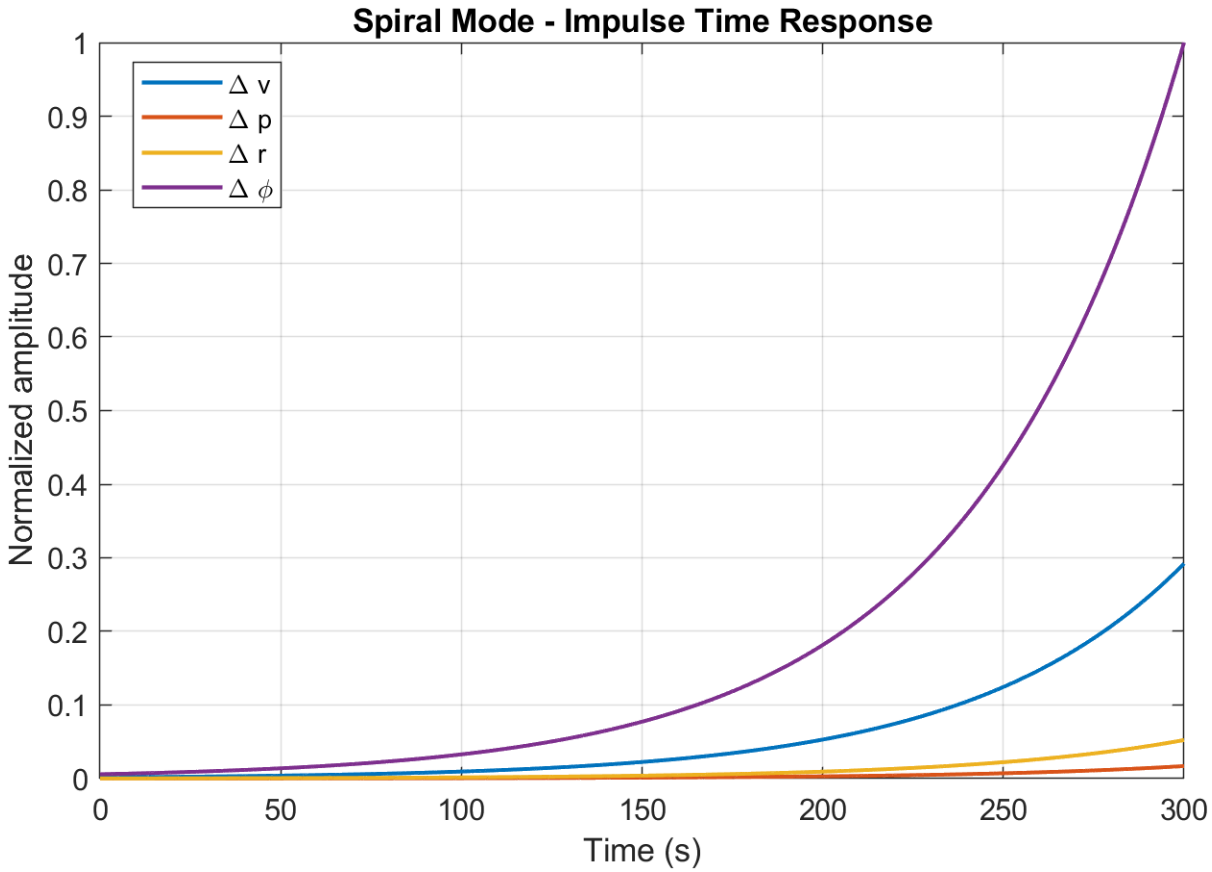


Figure 3.5: Spiral Mode – Impulse Time Response (normalized).

The bank angle Δφ (purple) dominates and grows exponentially (very slowly) over the 300-second window, confirming the slightly unstable spiral mode ($\lambda = +0.017$, $T_2 = 40.6$ s). The sideslip Δv follows the same divergent trend, while Δp and Δr stay small. This slow divergence is the characteristic signature of spiral instability: the aircraft gradually rolls into a banked turn if left uncorrected, but the time scale is slow enough that a pilot corrects it easily.

3.5.6 Global Free Responses

In addition to the individual mode plots, the free response of the full 4×4 systems is simulated (using ode45) with a nonzero initial condition on all four state variables:

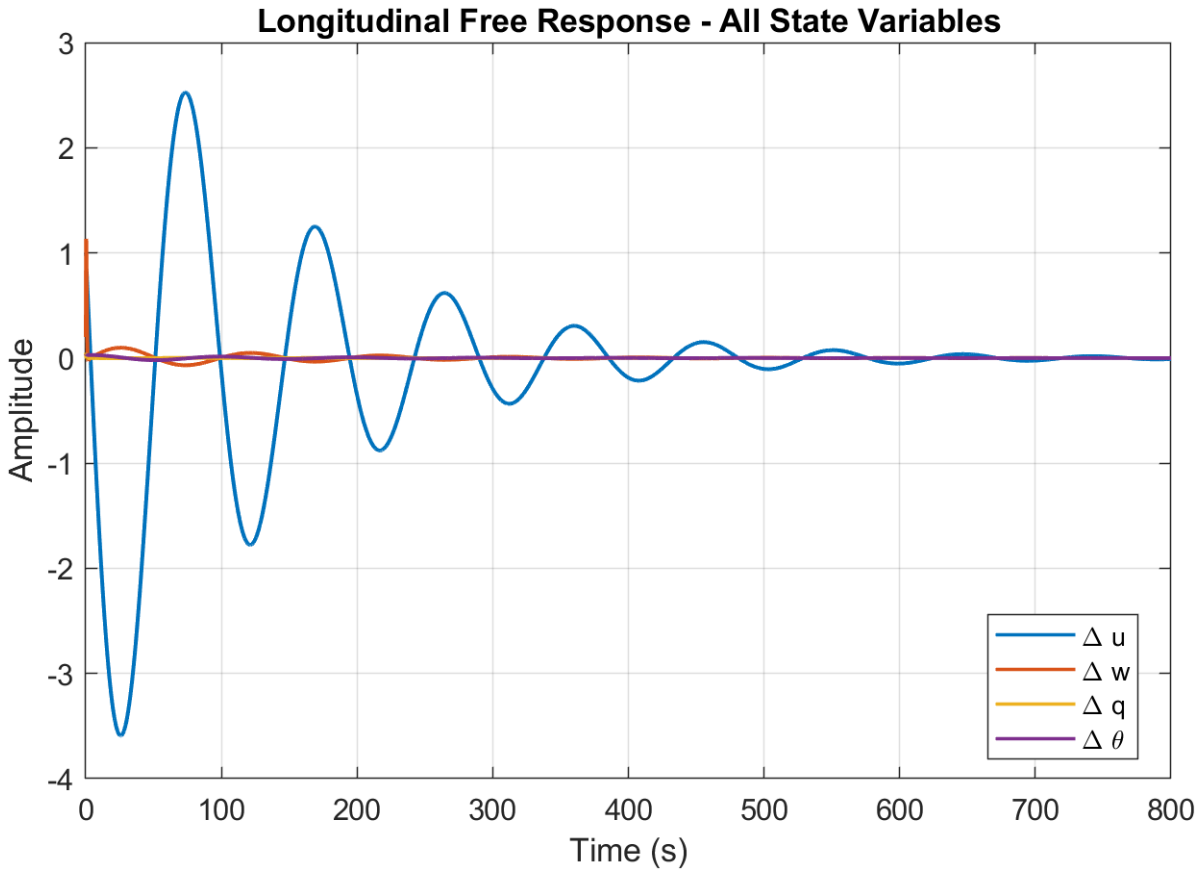


Figure 3.6: Longitudinal free response with mixed initial conditions.

In the longitudinal response, after the fast short-period transient dies out (within ~5 s), the phugoid oscillation in Δu (blue) dominates the long-term response for nearly 800 seconds, with slow damping characteristic of $\xi = 0.11$.

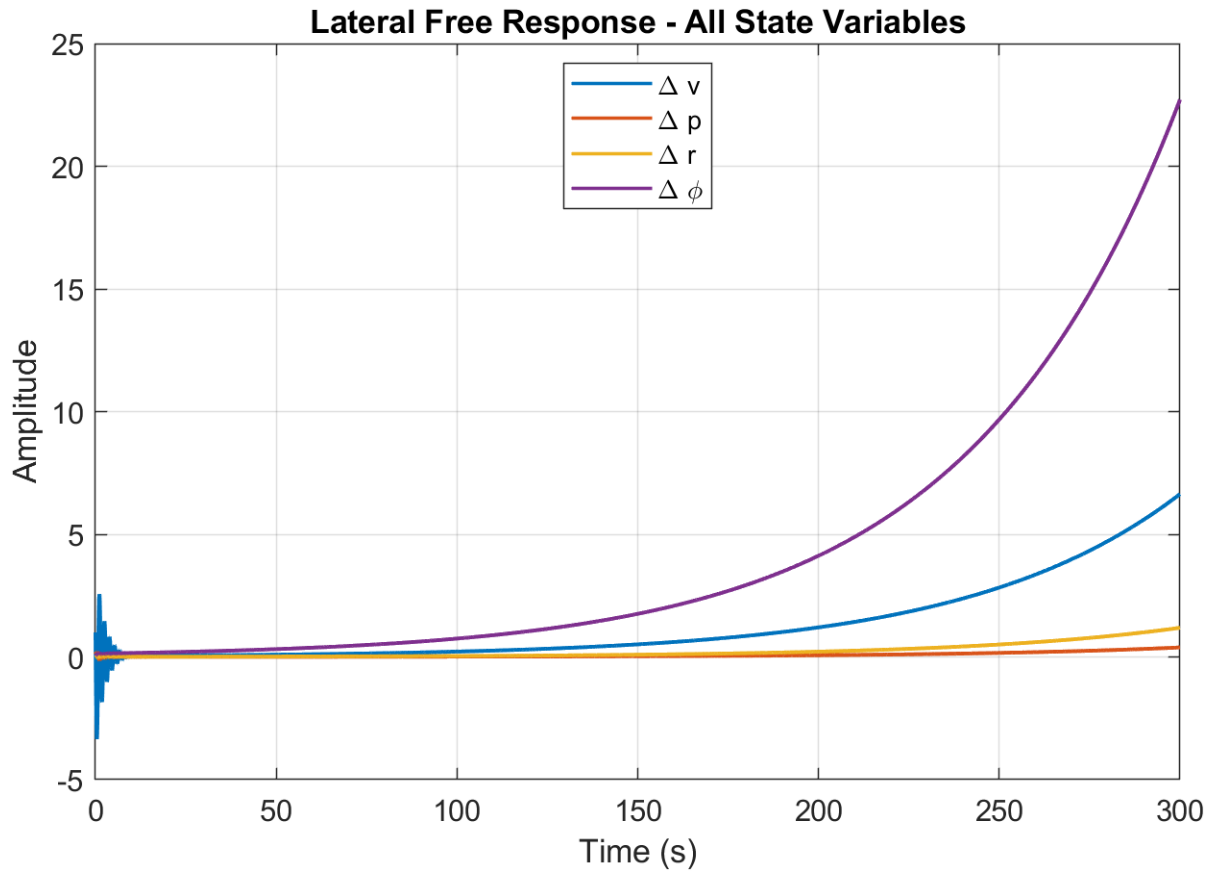


Figure 3.7: Lateral free response with mixed initial conditions.

In the lateral response, a brief Dutch roll transient is visible in the first few seconds, followed by the slow spiral divergence in Δφ (purple) and Δv (blue) over the 300-second window. This is the combined physical picture of the three lateral modes superimposed.

3.6 Control Transfer Functions

The control transfer functions are computed from the state-space matrices using $X(s)/\eta(s) = (sI - A)^{-1} \cdot B$. The common denominators are the longitudinal and lateral characteristic polynomials:

$$\Delta_Long(s) = s^4 + 5.662s^3 + 12.075s^2 + 0.201s + 0.053$$

$$\Delta_Lat(s) = s^4 + 8.387s^3 + 26.629s^2 + 154.426s - 2.642$$

3.6.1 Bode Diagrams

The frequency response of the four key transfer functions is computed numerically by evaluating $(sI - A)^{-1} \cdot B$ at $s = j\omega$ across a logarithmic frequency range:

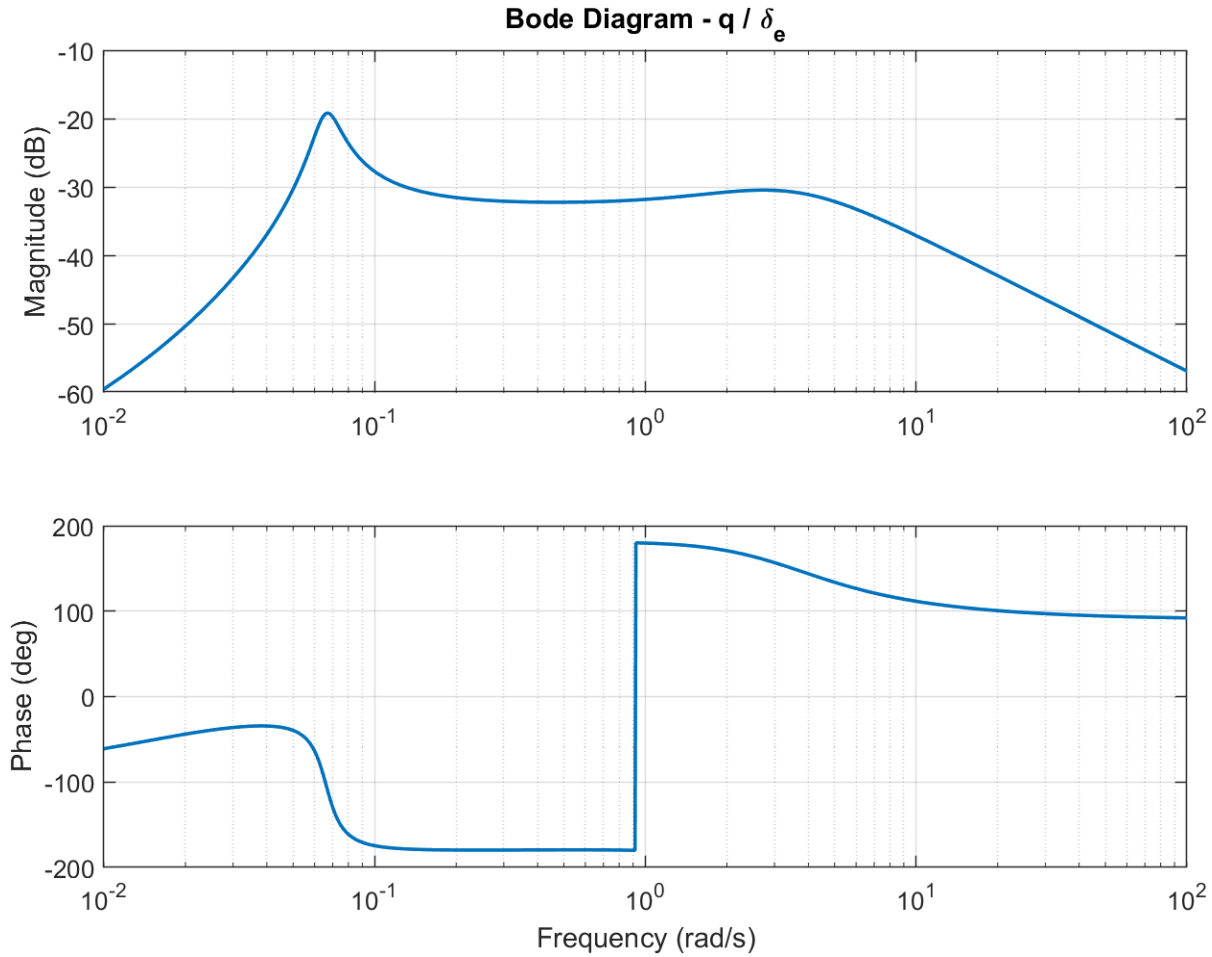


Figure 3.8: Bode diagram of q/δe (pitch rate response to elevator).

The pitch rate response q/δe shows two resonance peaks in magnitude, located at the phugoid frequency (~0.07 rad/s) and near the short-period frequency (~3.5 rad/s), with a flat intermediate plateau. The phase drops by ~360° across the full frequency range, characteristic of a 4th-order system with a numerator of degree 3.

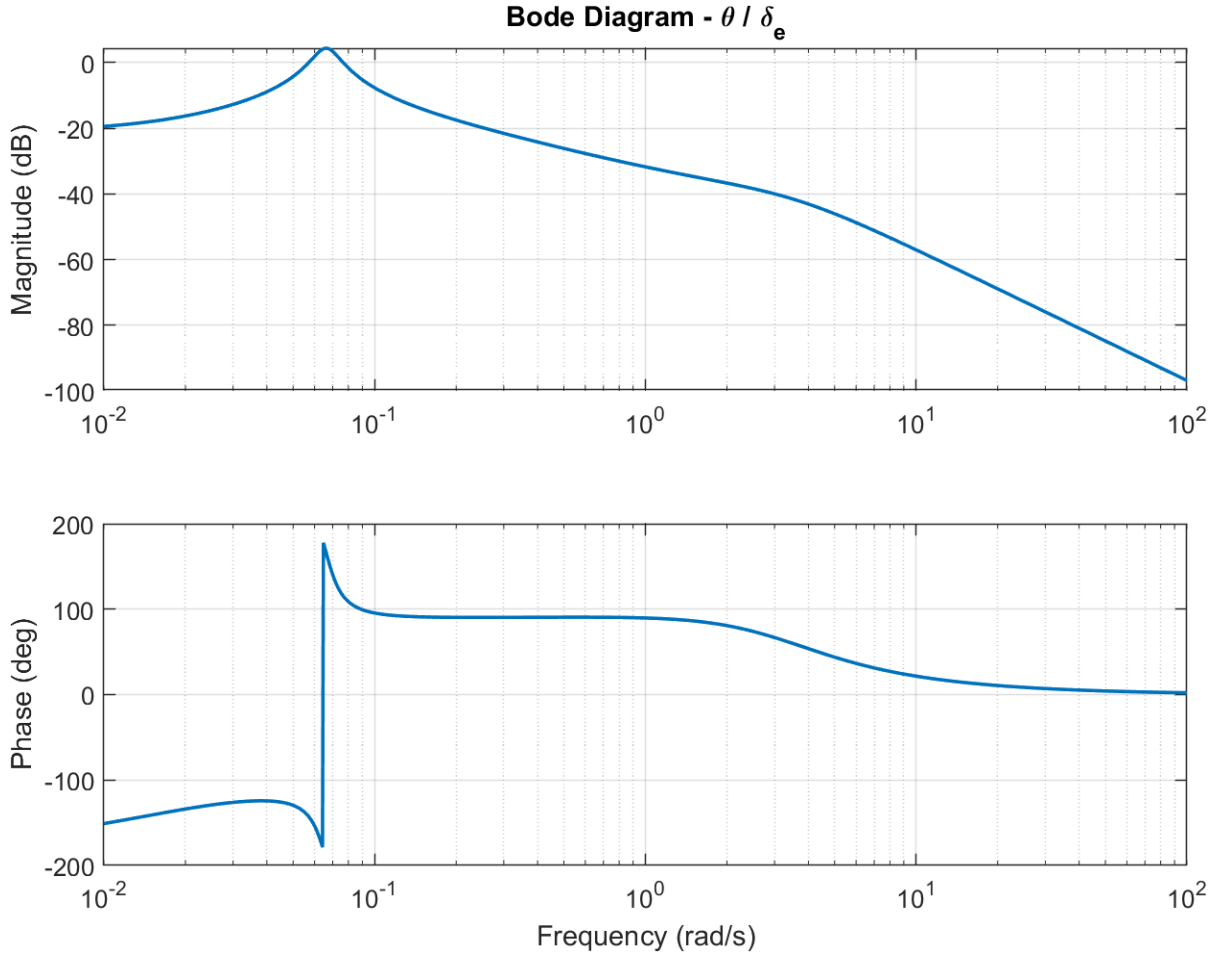


Figure 3.9: Bode diagram of θ/δe (pitch attitude response to elevator).

The pitch attitude response θ/δe is dominated by the phugoid resonance at low frequency, with a high DC gain and a −40 dB/decade roll-off at higher frequencies. This is the primary transfer function used for pitch autopilot and attitude hold design.

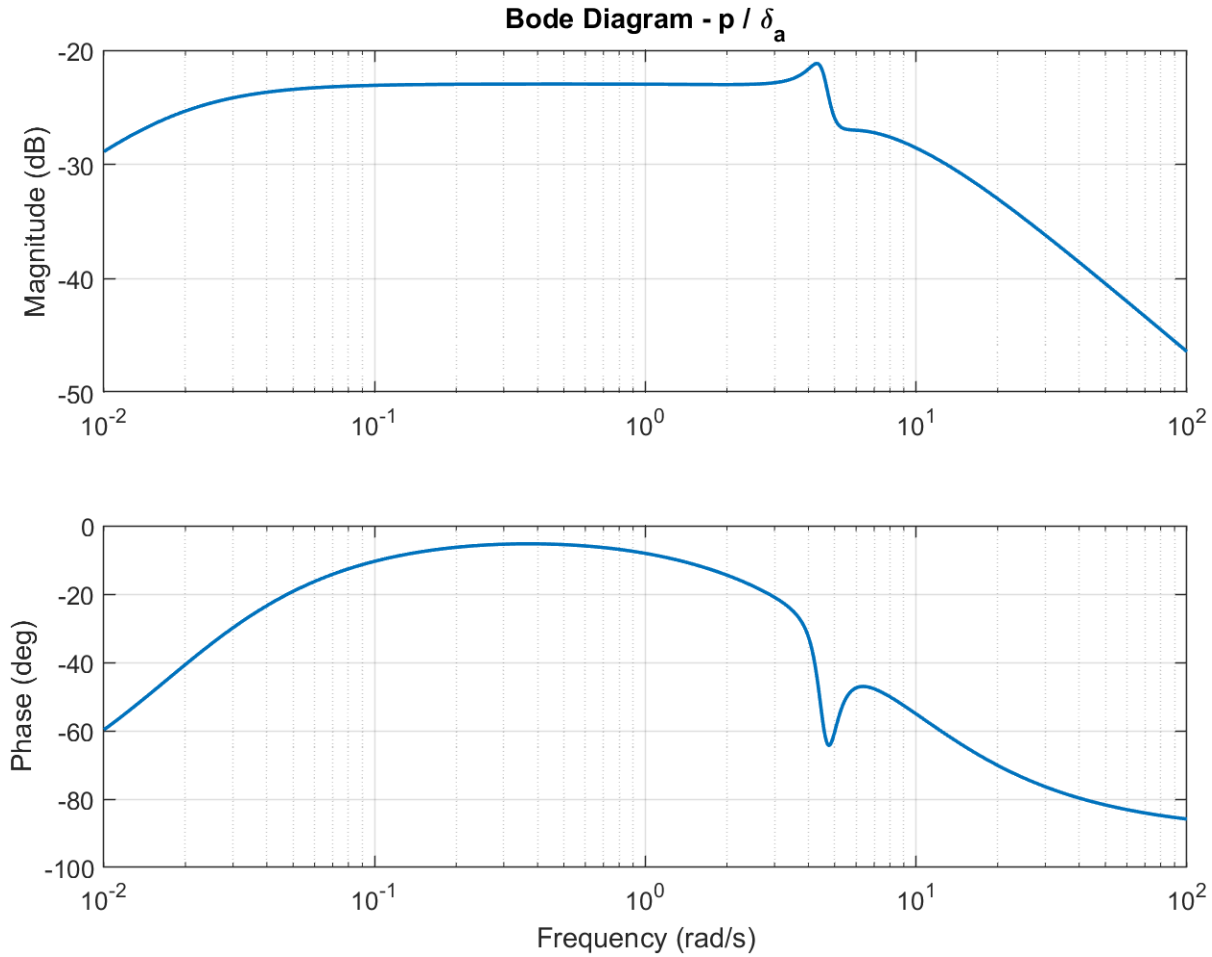


Figure 3.10: Bode diagram of p/δa (roll rate response to aileron).

The roll rate response p/δa has the broadest bandwidth of the four transfer functions, reflecting the aileron's role as the primary roll control. The magnitude is flat from DC up to ~5 rad/s, with a small peak at the Dutch roll frequency before the high-frequency roll-off.

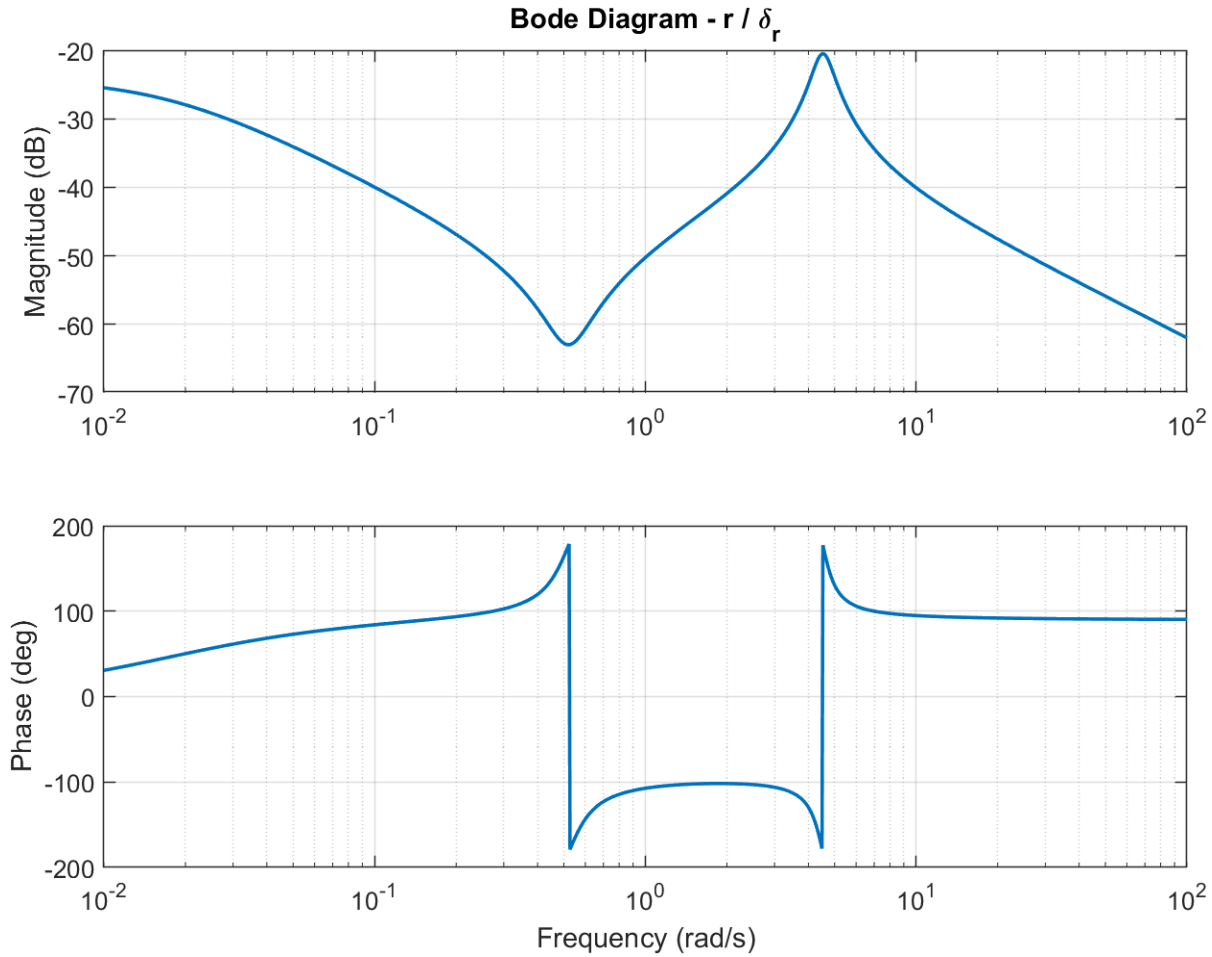


Figure 3.11: Bode diagram of r/δr (yaw rate response to rudder).

The yaw rate response r/δr shows a clear Dutch roll resonance peak at ω ≈ 4.5 rad/s and a notch at low frequency caused by a numerator zero near the spiral pole. This is the key transfer function used for yaw damper design.

3.6.2 Root Locus

The root locus of the four main transfer functions (q/δe, θ/δe, p/δa, r/δr) is plotted on a single s-plane with iso-damping and iso-natural-frequency grid lines:

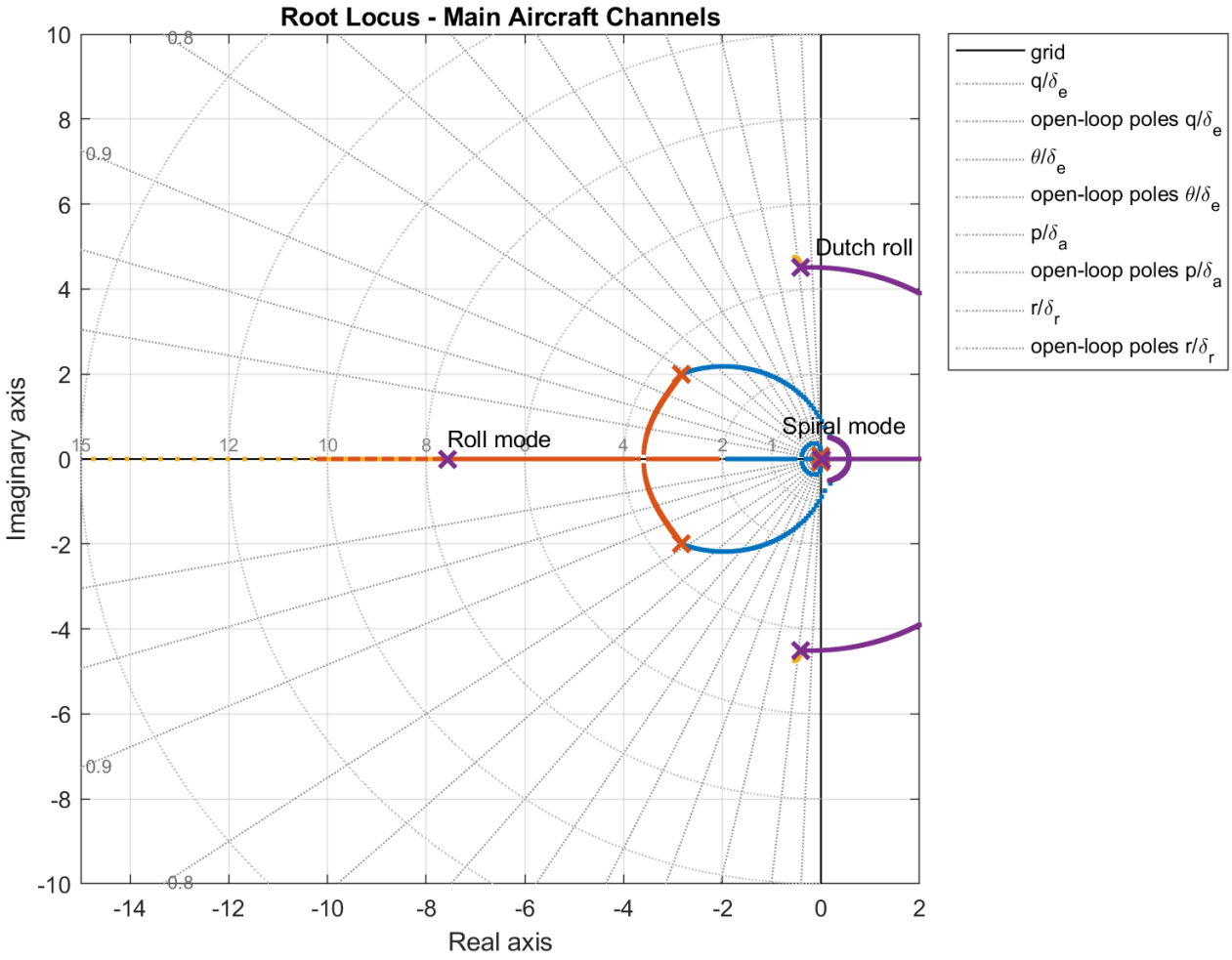


Figure 3.12: Root locus of the main transfer functions with s-plane grid.

The open-loop poles of the lateral system are labeled directly on the s-plane: the Roll mode at $\lambda \approx -7.6$ (far left on the real axis), the Dutch Roll pair near $-0.4 \pm 4.5j$, and the slightly unstable Spiral mode at $+0.017$ (just right of the imaginary axis). The locus branches show how the closed-loop poles migrate as the feedback gain increases on each transfer function. This visualization directly supports the SAS design choices made in Section 3.8.

3.7 Flying Quality Assessment

Aircraft classification:

- Class I: small, light airplane (m = 1816 kg)
- Flight Phase: Category B (cruise at 7620 m, M = 0.6)

3.7.1 Short Period Mode Assessment

Damping ratio criterion (Cat B):

Level 1 requirement: $0.30 \leq \xi_SP \leq 2.00$. Result: $\xi_SP = 0.8155 \rightarrow$ Level 1 SATISFIED.

Control Anticipation Parameter (CAP):

$$n/\alpha = (\bar{q} \cdot S \cdot C_L\alpha) / (m \cdot g) = 37.04\ g/rad$$

$$CAP = \omega^2_nSP / (n/\alpha) = 3.4623^2 / 37.04 = 0.324$$

Level 1 requirement (Cat B, Class I): 0.085 ≤ CAP ≤ 3.6. Result: CAP = 0.324 → Level 1 SATISFIED.

Short period overall assessment: Level 1.

3.7.2 Phugoid Mode Assessment

Level 1 requirement: ξ_Ph ≥ 0.04. Result: ξ_Ph = 0.1113 → Level 1 SATISFIED.

3.7.3 Roll Mode Assessment

Level 1 requirement (Cat B): T_R ≤ 1.4 s. Result: T_Roll = 0.132 s → Level 1 SATISFIED.

3.7.4 Spiral Mode Assessment

The spiral mode is unstable (λ = +0.0171). Level 1 requirement for Category B: T_2 ≥ 20 s. Result: T_2 = 40.6 s → Level 1 SATISFIED.

3.7.5 Dutch Roll Mode Assessment

Level 1 requirements (Cat B): ξ ≥ 0.08, ξ·ω_n ≥ 0.15, ω_n ≥ 0.5 rad/s.

- ξ_DR = 0.0922 ≥ 0.08 → SATISFIED
- ξ·ω_n = 0.4169 ≥ 0.15 → SATISFIED
- ω_n_DR = 4.5233 ≥ 0.5 → SATISFIED

Dutch roll overall assessment: Level 1.

3.7.6 Flying Quality Summary

Mode	Key Parameter	Value	Requirement	Level
Short Period	ξ / CAP	0.816 / 0.324	0.30–2.00 / 0.085–3.6	Level 1
Phugoid	ξ	0.111	≥ 0.04	Level 1
Roll	T_Roll	0.132 s	≤ 1.4 s	Level 1
Spiral	T_2	40.6 s	≥ 20 s	Level 1
Dutch Roll	ξ / ξω_n / ω_n	0.092 / 0.417 / 4.52	0.08 / 0.15 / 0.5	Level 1

All five stability modes of the C2 aircraft satisfy Level 1 flying quality requirements per MIL-F-8785C for a Class I airplane in Category B flight phase. The aircraft possesses

satisfactory flying qualities with no mandatory SAS, although the Short Period and Dutch Roll damping margins are relatively modest and can be improved through SAS design.

3.8 Stability Augmentation System (SAS) Design

Although all modes satisfy Level 1, a Pitch Damper and a Yaw Damper are designed to improve the damping margins of the Short Period and Dutch Roll modes respectively. The design uses the pole placement method: given target eigenvalues, the feedback gain vector K is computed so that the augmented system matrix $A^* = A - B \cdot K^T$ has the desired eigenvalues.

3.8.1 Pitch Damper (Short Period Enhancement)

Objective: increase the short period damping ratio from $\xi_{SP} = 0.816$ to $\xi_{SP} = 0.95$, while keeping the natural frequency ω_{n_SP} and the phugoid eigenvalues unchanged.

Target poles (longitudinal, 4 eigenvalues):

- $\lambda_{SP_new} = -3.2892 \pm 1.0811j$ (new short period)
- $\lambda_{Ph} = -0.0074 \pm 0.0658j$ (unchanged phugoid)

The controllability matrix $V = [B, AB, A^2B, A^3B]$ has full rank 4, so the system is fully controllable from the elevator input.

SAS gain (pole placement):

$$K_{long} = [0.0048,\ 0.0682,\ -6.4794,\ 0.0093]$$

The dominant term is the pitch rate feedback ($k_3 = -6.48$), which is the classical pitch damper architecture. After augmentation, the Short Period eigenvalues are $-3.2892 \pm 1.0811j$, giving $\omega_{n_SP} = 3.46$ rad/s and $\xi_{SP} = 0.95$ (target achieved). The phugoid eigenvalues are preserved as designed.

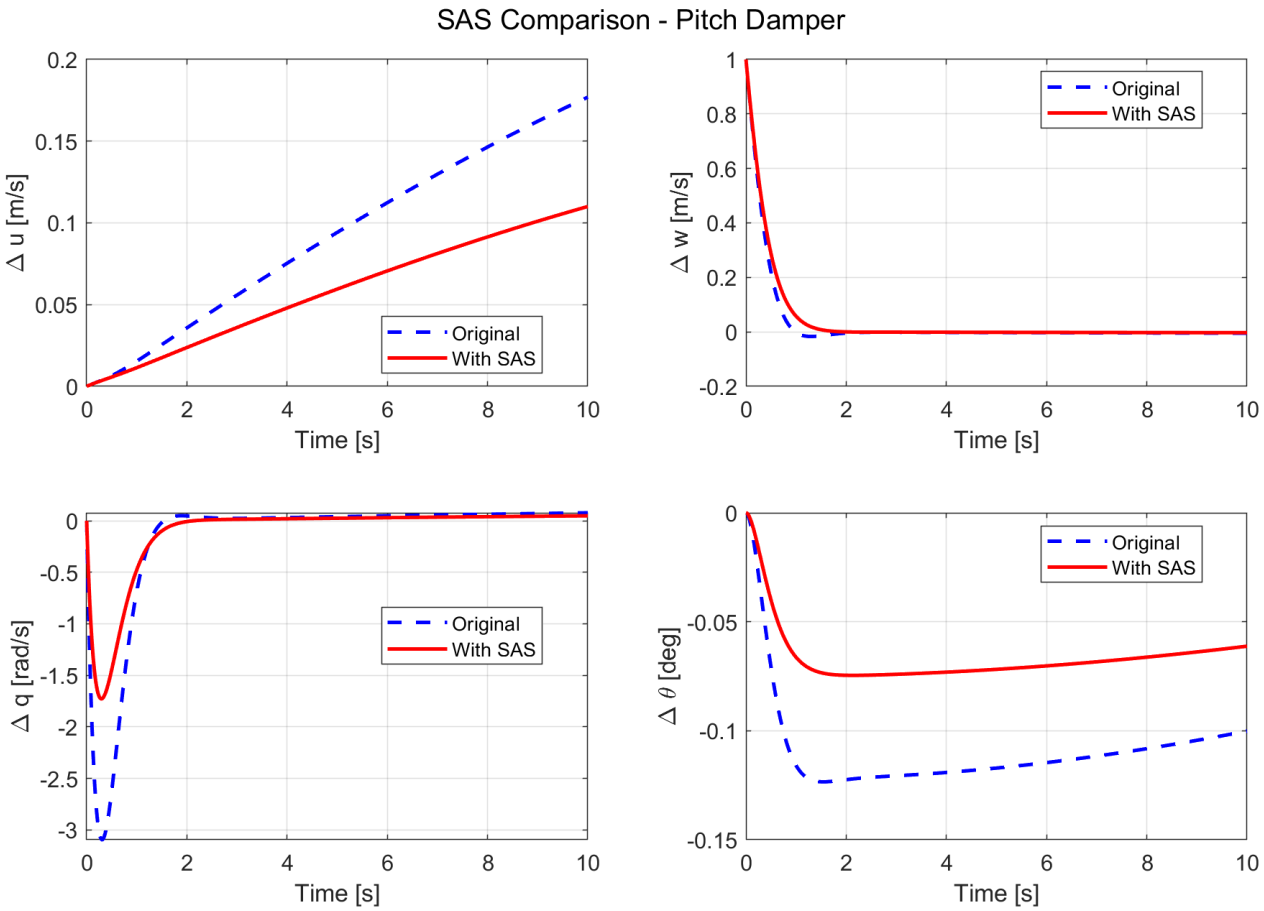


Figure 3.13: SAS comparison – Pitch Damper (original vs augmented).

For a unit initial perturbation in Δw, the original Δq peak reaches −3 rad/s (blue dashed), while with SAS it is limited to −1.8 rad/s (red solid), a 40% reduction. Similarly, the Δθ peak is reduced from −0.125° to −0.075°. The phugoid-related slow drift in Δu remains visible in both responses since the phugoid was intentionally left unchanged. The pitch damper clearly improves the short-period transient behavior as targeted.

3.8.2 Yaw Damper (Dutch Roll Enhancement)

Objective: increase the Dutch roll damping ratio from ξ_DR = 0.092 to ξ_DR = 0.20, while keeping the natural frequency ω_n_DR and the Roll and Spiral eigenvalues unchanged.

Target poles (lateral, 4 eigenvalues):

- λ_DR_new = −0.9047 ± 4.4319j (new Dutch roll)
- λ_Roll = −7.5700 (unchanged)
- λ_Spiral = +0.0171 (unchanged)

The lateral controllability matrix has full rank 4 from the rudder input, so the system is fully controllable.

SAS gain (pole placement):

$$K_lat = [-0.0110,\ 0.0859,\ -12.022,\ 0.6319]$$

As expected for a yaw damper, the dominant term is the yaw rate feedback ($k_3 = -12.02$), which generates a counter-rudder deflection proportional to Δr that opposes the Dutch roll oscillation. After augmentation, the Dutch Roll eigenvalues are $-0.9047 \pm 4.4319j$, giving $\omega_n_DR = 4.52$ rad/s and $\xi_DR = 0.20$ (target achieved). The product $\xi \cdot \omega_n$ jumps from 0.417 to 0.905 (117% improvement). The Roll and Spiral modes are unchanged.

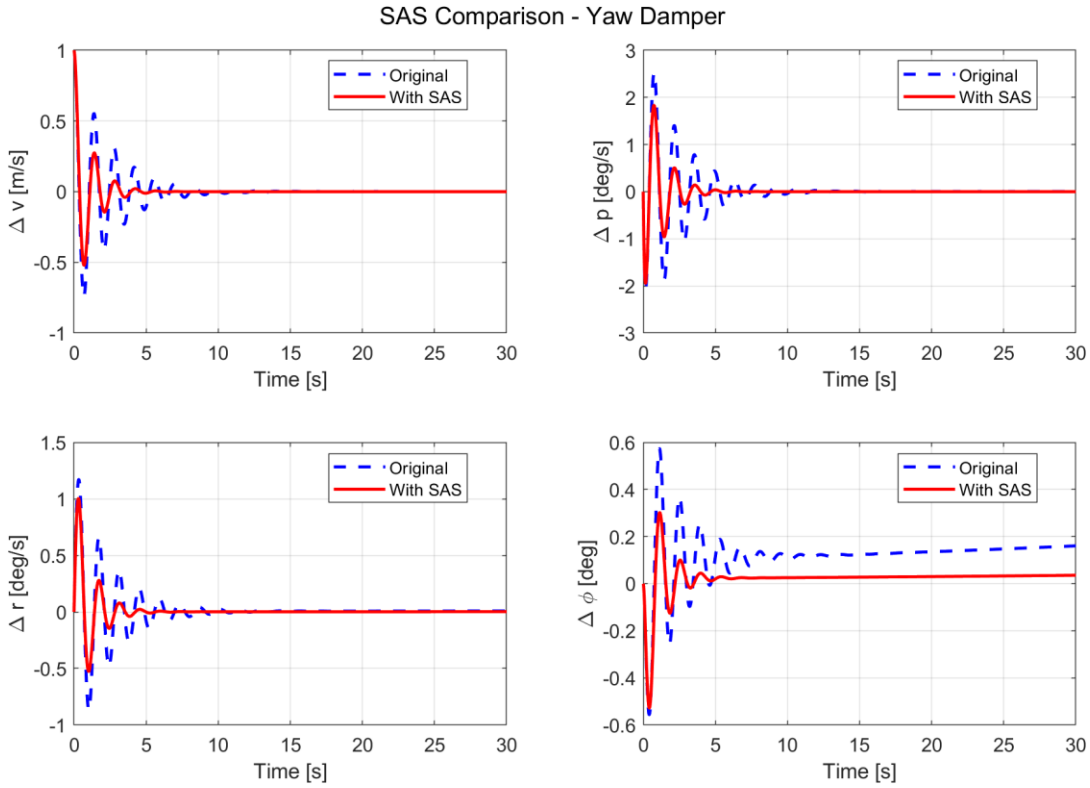


Figure 3.14: SAS comparison – Yaw Damper (original vs augmented).

For a unit sideslip perturbation, the original Dutch roll oscillation persists for ~15 seconds (blue dashed), while with SAS (red solid) it is essentially damped within 5 seconds. The amplitude reduction is clearly visible in Δv, Δp, Δr. In the Δφ subplot, the original response shows a residual bank angle of ~0.15° at t = 30 s (due to the spiral divergence), while the augmented response shows only ~0.04°. The SAS has eliminated the Dutch roll contribution but not the spiral divergence (which was intentionally kept unchanged).

3.9 Conclusion

The dynamic stability analysis of the C2 aircraft has been carried out end-to-end using MATLAB. The main findings are:

- The state-space matrices A_Long, B_Long, A_Lat, B_Lat were computed from the aircraft data, applying the I_n inversion to account for the I_xz inertial coupling (since I_xz/I_xx = 0.25 is not negligible).
- The longitudinal eigenvalues yield a well-damped Short Period mode (ω_n = 3.46 rad/s, ξ = 0.82) and a lightly damped Phugoid mode (ω_n = 0.066 rad/s, ξ = 0.11).
- The lateral eigenvalues yield a fast Roll mode (T = 0.13 s), a slightly unstable Spiral mode (T_2 = 40.6 s), and a lightly damped Dutch Roll mode (ω_n = 4.52 rad/s, ξ = 0.09).
- All five modes satisfy Level 1 flying quality requirements per MIL-F-8785C for a Class I aircraft in Category B flight phase.
- Although no SAS is strictly required, a Pitch Damper and a Yaw Damper were designed by pole placement to improve the Short Period damping (ξ: 0.82 → 0.95) and the Dutch Roll damping (ξ: 0.09 → 0.20), with clear improvements visible in the time response comparisons.

The C2 aircraft is therefore a satisfactory airplane from a flying qualities standpoint, safe to fly without stability augmentation but benefiting from a classical pitch/yaw damper combination to improve pilot comfort and ride quality.

General Conclusion

This report has presented a complete dynamic stability and flying qualities analysis of the C2 aircraft, conducted in accordance with the MIL-F-8785C military specification.

The theoretical framework developed in Chapters 1 and 2 established the mathematical basis for the analysis: the linearized equations of motion, the state-space representation, and the flying quality requirements applicable to a Class I aircraft in Category B flight phase.

The numerical application in Chapter 3 demonstrated that the C2 aircraft possesses satisfactory flying qualities across all five stability modes. The longitudinal dynamics are characterized by a well-damped Short Period mode ($\omega_n = 3.46$ rad/s, $\xi = 0.82$, CAP = 0.32) and a lightly damped but stable Phugoid mode ($\xi = 0.11$, $T \approx 95$ s), both satisfying Level 1 requirements. The lateral-directional dynamics show a fast Roll mode ($T = 0.13$ s), a mildly unstable but acceptable Spiral mode ($T_2 = 40.6$ s), and a Dutch Roll mode ($\omega_n = 4.52$ rad/s, $\xi = 0.09$) meeting all three Level 1 criteria for Category B.

Although no Stability Augmentation System is strictly required, a Pitch Damper and a Yaw Damper were designed by pole placement to improve the damping margins of the Short Period (ξ: 0.82 → 0.95) and Dutch Roll (ξ: 0.09 → 0.20) modes respectively, confirming the full controllability of the system.

Overall, the C2 aircraft can be considered a dynamically well-behaved platform, safe to operate without augmentation and capable of benefiting from a classical pitch/yaw damper combination to further improve pilot comfort and handling precision.

Appendix A: MATLAB Code

The complete MATLAB code :

```
%% ========================================================================= %  C2 AIRCRAFT - FLYING QUALITIES ANALYSIS % Course: Mecanique du vol - Qualites de vol (IPSA Aero4) %  Professor: Philippe Yazigi %% ========================================================================= clc; clear; close all;  outdir = fullfile(pwd, 'report_figures'); if ~exist(outdir, 'dir'), mkdir(outdir); end  %% ========================================================================= %  1. AIRCRAFT C2 DATA %% ========================================================================= g = 9.81;  % Geometry S = 12.63; cbar = 1.65; b = 8.02; AR = 5.09; e = 0.94;  % Flight condition h = 7620; Mach = 0.6; U0 = 185.9; qbar = 9500; rho = 0.55; theta0 = 0*pi/180;  % Mass / inertia m = 1816; Ixx = 1085; Iyy = 6511; Izz = 7053; Ixz = 271;  % Steady-state CL0 = 0.15; CD0 = 0.022;  % Longitudinal stability derivatives CDu = 0; CDa = 0.12; CLu = 0.084; CLa = 5.5; CLadot = 4.2; CLq = 10; Cmu = 0; Cma = -0.24; Cmadot = -9.6; Cmq = -17.7;  % Lateral stability derivatives Clb = -0.11; Clp = -0.39; Clr = 0.28; Cnb = 0.17;  Cnp = 0.09;  Cnr = -0.26; Cyb = -1;    Cyp = -0.14; Cyr = 0.61;  % Control derivatives CDde = 0; CLde = 0.038; Cmde = -0.88; Cyda = 0; Clda = 0.1;   Cnda = -0.003; Cydr = 0.28; Cldr = 0.05; Cndr = -0.12;  %% ========================================================================= %  2. DIMENSIONAL LONGITUDINAL DERIVATIVES %% ========================================================================= Xu = -(qbar*S)/(m*U0) * (2*CD0 + CDu); Zu = -(qbar*S)/(m*U0) * (2*CL0 + CLu); Mu =  (qbar*S*cbar)/(Iyy*U0) * Cmu;
Xw = (qbar*S)/(m*U0) * CL0 * (1 - (2/(pi*e*AR))*CLa); Zw = -(qbar*S)/(m*U0) * (CD0 + CLa); Mw = (qbar*S*cbar)/(Iyy*U0) * Cma;  Zwdot = (qbar*S*cbar)/(2*m*U0^2) * (2*CD0 + CLadot); Mwdot = (qbar*S*cbar^2)/(2*Iyy*U0^2) * Cmadot;  Zq = (qbar*S*cbar)/(2*m*U0) * CLq; Mq = (qbar*S*cbar^2)/(2*Iyy*U0) * Cmq;  % Longitudinal control derivatives Xde = (qbar*S)/(m*U0) * CDde; Zde = (qbar*S)/(m*U0) * CLde; Mde = (qbar*S*cbar)/(Iyy*U0) * Cmde; XdT = 0; ZdT = 0; MdT = 0;  %% ========================================================================= %  3. LONGITUDINAL MATRICES %  States: [du dw dq dtheta] %% ========================================================================= ALong = [ Xu,  Xw,  0,              -g*cos(theta0);           Zu,  Zw,  U0,             -g*sin(theta0);           Mu + Zu*Mwdot,  Mw + Zw*Mwdot,  Mq + U0*Mwdot,  0;           0,   0,   1,              0 ];  BLong = [ Xde,  XdT;           Zde,  ZdT;           Mde + Zde*Mwdot,  MdT + ZdT*Mwdot;           0,    0 ];  disp(' ') disp('================================') disp('LONGITUDINAL RESULTS') disp('================================') disp('ALong = '); disp(ALong) disp('BLong = '); disp(BLong)  %% ========================================================================= %  4. DIMENSIONAL LATERAL DERIVATIVES (convention prof avec matrice In) %% ========================================================================= Yv = (qbar*S)/(m*U0) * Cyb; Yp = (qbar*S*b)/(2*m*U0) * Cyp; Yr = (qbar*S*b)/(2*m*U0) * Cyr;  Lv = (qbar*S*b)/(Ixx*U0) * Clb; Lp = (qbar*S*b^2)/(2*Ixx*U0) * Clp; Lr = (qbar*S*b^2)/(2*Ixx*U0) * Clr;  Nv = (qbar*S*b)/(Izz*U0) * Cnb; Np = (qbar*S*b^2)/(2*Izz*U0) * Cnp;
Nr = (qbar*S*b^2)/(2*Izz*U0) * Cnr;  Yda = (qbar*S)/(m*U0) * Cyda; Ydr = (qbar*S)/(m*U0) * Cydr; Lda = (qbar*S*b)/(Ixx*U0) * Clda; Ldr = (qbar*S*b)/(Ixx*U0) * Cldr; Nda = (qbar*S*b)/(Izz*U0) * Cnda; Ndr = (qbar*S*b)/(Izz*U0) * Cndr;  %% ========================================================================= %  5. LATERAL MATRICES (with Ixz coupling via InLat inversion) %% ========================================================================= AnLat = [ Yv,  Yp,  Yr - U0,              g*cos(theta0);           Lv,  Lp,  Lr,       0;           Nv,  Np,  Nr,       0;           0,   1,   0,        0 ];  BnLat = [ Ydr,  Yda;           Ldr,  Lda;           Ndr,  Nda;           0,    0 ];  InLat = [1, 0,          0,        0;          0, 1,   -Ixz/Ixx,       0;          0, -Ixz/Izz,    1,       0;          0, 0,           0,       1];  ALat = InLat \ AnLat; BLat = InLat \ BnLat;  disp(' ') disp('================================') disp('      LATERAL RESULTS') disp('================================') disp('ALat = '); disp(ALat) disp('BLat = '); disp(BLat)  %% ========================================================================= %  6. EIGENVALUE ANALYSIS - LONGITUDINAL %% ========================================================================= [Vlong, Dlong] = eig(ALong); lambda_long = diag(Dlong); char_poly_long = poly(ALong);  disp(' ') disp('Longitudinal eigenvalues:'); disp(lambda_long) disp('Longitudinal eigenvectors:'); disp(Vlong) disp('Longitudinal characteristic polynomial coefficients:'); disp(char_poly_long)  [~, idxL] = sort(abs(imag(lambda_long)), 'descend');
idxSP = idxL(1:2); idxPH = idxL(3:4);  lambdaSP = lambda_long(idxSP); lambdaPH = lambda_long(idxPH);  sigma_SP = real(lambdaSP(1)); omega_d_SP = abs(imag(lambdaSP(1))); omega_n_SP = sqrt(sigma_SP^2 + omega_d_SP^2); zeta_SP = -sigma_SP / omega_n_SP;  sigma_PH = real(lambdaPH(1)); omega_d_PH = abs(imag(lambdaPH(1))); omega_n_PH = sqrt(sigma_PH^2 + omega_d_PH^2); zeta_PH = -sigma_PH / omega_n_PH;  disp(' '); disp('--- SHORT PERIOD ---') disp(['lambda_SP = ', num2str(lambdaSP(1))]) disp(['omega_n_SP = ', num2str(omega_n_SP), ' rad/s']) disp(['zeta_SP = ', num2str(zeta_SP)])
disp(' '); disp('--- PHUGOID ---') disp(['lambda_PH = ', num2str(lambdaPH(1))]) disp(['omega_n_PH = ', num2str(omega_n_PH), ' rad/s']) disp(['zeta_PH = ', num2str(zeta_PH)])  %% ========================================================================= %  7. EIGENVALUE ANALYSIS - LATERAL %% ========================================================================= [Vlat, Dlat] = eig(ALat); lambda_lat = diag(Dlat); char_poly_lat = poly(ALat);  disp(' ') disp('Lateral eigenvalues:'); disp(lambda_lat) disp('Lateral eigenvectors:'); disp(Vlat) disp('Lateral characteristic polynomial coefficients:'); disp(char_poly_lat)  tolImag = 1e-8;
idxReal = find(abs(imag(lambda_lat)) < tolImag); idxCplx = find(abs(imag(lambda_lat)) >= tolImag);  lambda_real = real(lambda_lat(idxReal)); [~, iRoll] = min(lambda_real); [~, iSpir] = min(abs(lambda_real));  idxRoll = idxReal(iRoll);
idxSpir = idxReal(iSpir);  lambda_roll = lambda_lat(idxRoll);
lambda_spiral = lambda_lat(idxSpir);  idxDR = idxCplx(1:2); lambdaDR_pair = lambda_lat(idxDR);  sigma_DR = real(lambdaDR_pair(1)); omega_d_DR = abs(imag(lambdaDR_pair(1))); omega_n_DR = sqrt(sigma_DR^2 + omega_d_DR^2); zeta_DR = -sigma_DR / omega_n_DR;  T_roll = -1/real(lambda_roll);  disp(' '); disp('--- ROLL MODE ---') disp(['lambda_roll = ', num2str(real(lambda_roll))]) disp(['T_roll = ', num2str(T_roll), ' s'])  disp(' '); disp('--- SPIRAL MODE ---')
disp(['lambda_spiral = ', num2str(real(lambda_spiral))]) if real(lambda_spiral) > 0     disp(['T2 (double) = ',
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num2str(log(2)/real(lambda_spiral)), ' s (UNSTABLE)']) end  disp(' '); disp('--- DUTCH ROLL ---') disp(['lambda_DR = ',
num2str(lambdaDR_pair(1))]) disp(['omega_n_DR = ', num2str(omega_n_DR), ' rad/s']) disp(['zeta_DR = ', num2str(zeta_DR)])
disp(['zeta*omega_n = ', num2str(zeta_DR*omega_n_DR)])  %%
======================================================================== %  8. LONGITUDINAL MODE SHAPES (NORMALIZED) %%
======================================================================== VSP = Vlong(:, idxSP); VPH = Vlong(:, idxPH);
tSP = linspace(0, 20, 2000); tPH = linspace(0, 800, 4000);  du_SP     = real(VSP(1,1)*exp(lambdaSP(1)*tSP) +
VSP(1,2)*exp(lambdaSP(2)*tSP)); dw_SP     = real(VSP(2,1)*exp(lambdaSP(1)*tSP) + VSP(2,2)*exp(lambdaSP(2)*tSP)); dq_SP
= real(VSP(3,1)*exp(lambdaSP(1)*tSP) + VSP(3,2)*exp(lambdaSP(2)*tSP)); dtheta_SP = real(VSP(4,1)*exp(lambdaSP(1)*tSP) +
VSP(4,2)*exp(lambdaSP(2)*tSP));  du_PH     = real(VPH(1,1)*exp(lambdaPH(1)*tPH) + VPH(1,2)*exp(lambdaPH(2)*tPH)); dw_PH
= real(VPH(2,1)*exp(lambdaPH(1)*tPH) + VPH(2,2)*exp(lambdaPH(2)*tPH)); dq_PH     = real(VPH(3,1)*exp(lambdaPH(1)*tPH) +
VPH(3,2)*exp(lambdaPH(2)*tPH)); dtheta_PH = real(VPH(4,1)*exp(lambdaPH(1)*tPH) + VPH(4,2)*exp(lambdaPH(2)*tPH));  %
Normalize scale_SP = max([max(abs(du_SP)) max(abs(dw_SP)) max(abs(dq_SP)) max(abs(dtheta_SP))]) + eps; scale_PH =
max([max(abs(du_PH)) max(abs(dw_PH)) max(abs(dq_PH)) max(abs(dtheta_PH))]) + eps;
 du_SP = du_SP/scale_SP; dw_SP = dw_SP/scale_SP; dq_SP = dq_SP/scale_SP; dtheta_SP = dtheta_SP/scale_SP;  du_PH =
du_PH/scale_PH; dw_PH = dw_PH/scale_PH; dq_PH = dq_PH/scale_PH; dtheta_PH = dtheta_PH/scale_PH;  fig =
figure('Visible','on','Position',[100 100 700 450]); plot(tSP, du_SP, 'LineWidth', 1.5); hold on; plot(tSP, dw_SP,
'LineWidth', 1.5); plot(tSP, dq_SP, 'LineWidth', 1.5); plot(tSP, dtheta_SP, 'LineWidth', 1.5); grid on; xlabel('Time
(s)'); ylabel('Normalized amplitude'); title('Short Period Mode - Impulse Time Response'); legend('\Delta u', '\Delta w',
'\Delta q', '\Delta \theta', 'Location', 'best'); xlim([0 2.5]); set(gca,'FontSize',11); exportgraphics(fig,
fullfile(outdir,'fig_mode_ShortPeriod.png'), 'Resolution', 200);  fig = figure('Visible','on','Position',[100 100 700
450]); plot(tPH, du_PH, 'LineWidth', 1.5); hold on; plot(tPH, dw_PH, 'LineWidth', 1.5); plot(tPH, dq_PH, 'LineWidth',
1.5); plot(tPH, dtheta_PH, 'LineWidth', 1.5); grid on; xlabel('Time (s)'); ylabel('Normalized amplitude'); title('Phugoid
Mode - Impulse Time Response'); legend('\Delta u', '\Delta w', '\Delta q', '\Delta \theta', 'Location', 'best');
set(gca,'FontSize',11); exportgraphics(fig, fullfile(outdir,'fig_mode_Phugoid.png'), 'Resolution', 200);  %%
======================================================================== %  9. LATERAL MODE SHAPES (NORMALIZED) %%
======================================================================== VDR = Vlat(:, idxDR); VRoll = Vlat(:, idxRoll);
VSpir = Vlat(:, idxSpir);  tDR = linspace(0, 100, 3000); tRoll = linspace(0, 20, 2000); tSpir = linspace(0, 300, 3000);
dv_DR = real(VDR(1,1)*exp(lambdaDR_pair(1)*tDR) + VDR(1,2)*exp(lambdaDR_pair(2)*tDR)); dp_DR =
real(VDR(2,1)*exp(lambdaDR_pair(1)*tDR) + VDR(2,2)*exp(lambdaDR_pair(2)*tDR)); dr_DR =
real(VDR(3,1)*exp(lambdaDR_pair(1)*tDR) + VDR(3,2)*exp(lambdaDR_pair(2)*tDR)); dphi_DR =
real(VDR(4,1)*exp(lambdaDR_pair(1)*tDR) + VDR(4,2)*exp(lambdaDR_pair(2)*tDR));  dv_roll =
real(VRoll(1)*exp(lambda_roll*tRoll)); dp_roll = real(VRoll(2)*exp(lambda_roll*tRoll)); dr_roll =
real(VRoll(3)*exp(lambda_roll*tRoll)); dphi_roll = real(VRoll(4)*exp(lambda_roll*tRoll));
dv_sp = real(VSpir(1)*exp(lambda_spiral*tSpir)); dp_sp = real(VSpir(2)*exp(lambda_spiral*tSpir)); dr_sp =
real(VSpir(3)*exp(lambda_spiral*tSpir)); dphi_sp = real(VSpir(4)*exp(lambda_spiral*tSpir));  scale_DR =
max([max(abs(dv_DR)) max(abs(dp_DR)) max(abs(dr_DR)) max(abs(dphi_DR))]) + eps; scale_roll = max([max(abs(dv_roll))
max(abs(dp_roll)) max(abs(dr_roll)) max(abs(dphi_roll))]) + eps; scale_sp = max([max(abs(dv_sp)) max(abs(dp_sp))
max(abs(dr_sp)) max(abs(dphi_sp))]) + eps;  dv_DR = dv_DR/scale_DR; dp_DR = dp_DR/scale_DR; dr_DR = dr_DR/scale_DR;
dphi_DR = dphi_DR/scale_DR;  dv_roll = dv_roll/scale_roll; dp_roll = dp_roll/scale_roll; dr_roll = dr_roll/scale_roll;
dphi_roll = dphi_roll/scale_roll;  dv_sp = dv_sp/scale_sp; dp_sp = dp_sp/scale_sp; dr_sp = dr_sp/scale_sp; dphi_sp =
dphi_sp/scale_sp;  fig = figure('Visible','on','Position',[100 100 700 450]); plot(tDR, dv_DR, 'LineWidth', 1.5); hold on;
plot(tDR, dp_DR, 'LineWidth', 1.5); plot(tDR, dr_DR, 'LineWidth', 1.5); plot(tDR, dphi_DR, 'LineWidth', 1.5); grid on;
xlabel('Time (s)'); ylabel('Normalized amplitude'); title('Dutch Roll Mode - Impulse Time Response'); legend('\Delta v',
'\Delta p', '\Delta r', '\Delta \phi', 'Location', 'best'); xlim([0 15]); set(gca,'FontSize',11); exportgraphics(fig,
fullfile(outdir,'fig_mode_DutchRoll.png'), 'Resolution', 200);  fig = figure('Visible','on','Position',[100 100 700 450]);
plot(tRoll, dv_roll, 'LineWidth', 1.5); hold on; plot(tRoll, dp_roll, 'LineWidth', 1.5); plot(tRoll, dr_roll, 'LineWidth',
1.5); plot(tRoll, dphi_roll, 'LineWidth', 1.5); grid on; xlabel('Time (s)'); ylabel('Normalized amplitude'); title('Roll
Mode - Impulse Time Response'); legend('\Delta v', '\Delta p', '\Delta r', '\Delta \phi', 'Location', 'best'); xlim([0
1]); set(gca,'FontSize',11); exportgraphics(fig, fullfile(outdir,'fig_mode_Roll.png'), 'Resolution', 200);  fig =
figure('Visible','on','Position',[100 100 700 450]); plot(tSpir, dv_sp, 'LineWidth', 1.5); hold on; plot(tSpir, dp_sp,
'LineWidth', 1.5); plot(tSpir, dr_sp, 'LineWidth', 1.5); plot(tSpir, dphi_sp, 'LineWidth', 1.5); grid on; xlabel('Time
(s)'); ylabel('Normalized amplitude'); title('Spiral Mode - Impulse Time Response'); legend('\Delta v', '\Delta p',
'\Delta r', '\Delta \phi', 'Location', 'best'); set(gca,'FontSize',11); exportgraphics(fig,
fullfile(outdir,'fig_mode_Spiral.png'), 'Resolution', 200);
 %% ======================================================================== %  10. GLOBAL FREE RESPONSES %%
======================================================================== x0_long = [1; 0.2; 0.05; 0.02]; [tt_long, x_long]
= ode45(@(t,x) ALong*x, linspace(0,800,4000), x0_long);  fig = figure('Visible','on','Position',[100 100 700 450]);
plot(tt_long, x_long(:,1), 'LineWidth', 1.3); hold on; plot(tt_long, x_long(:,2), 'LineWidth', 1.3); plot(tt_long,
x_long(:,3), 'LineWidth', 1.3); plot(tt_long, x_long(:,4), 'LineWidth', 1.3); grid on; xlabel('Time (s)');
ylabel('Amplitude'); title('Longitudinal Free Response - All State Variables'); legend('\Delta u', '\Delta w', '\Delta q',
'\Delta \theta', 'Location', 'best'); set(gca,'FontSize',11); exportgraphics(fig,
fullfile(outdir,'fig_free_longitudinal.png'), 'Resolution', 200);  x0_lat = [1; 0.2; 0.1; 0.05]; [tt_lat, x_lat] =
ode45(@(t,x) ALat*x, linspace(0,300,4000), x0_lat);  fig = figure('Visible','on','Position',[100 100 700 450]);
plot(tt_lat, x_lat(:,1), 'LineWidth', 1.3); hold on; plot(tt_lat, x_lat(:,2), 'LineWidth', 1.3); plot(tt_lat, x_lat(:,3),
'LineWidth', 1.3); plot(tt_lat, x_lat(:,4), 'LineWidth', 1.3); grid on; xlabel('Time (s)'); ylabel('Amplitude');
title('Lateral Free Response - All State Variables'); legend('\Delta v', '\Delta p', '\Delta r', '\Delta \phi',
'Location', 'best'); set(gca,'FontSize',11); exportgraphics(fig, fullfile(outdir,'fig_free_lateral.png'), 'Resolution',
200);  %% ======================================================================== %  11. FLYING QUALITY ASSESSMENT (Class
I, Cat B) %% ======================================================================== disp(' ')
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disp('============================================') disp('   FLYING QUALITY ASSESSMENT (Class I, Cat B)')
disp('============================================')  disp(['SP: omega_n=', num2str(omega_n_SP,'%.4f'), ', xi=',
num2str(zeta_SP,'%.4f')]) n_alpha = (qbar*S*CLa)/(m*g); CAP = omega_n_SP^2/n_alpha; disp(['    CAP = ',
num2str(CAP,'%.4f'), ' -> Level 1 (0.085-3.6)'])  disp(['Ph: omega_n=', num2str(omega_n_PH,'%.4f'), ', xi=',
num2str(zeta_PH,'%.4f'), ' -> Level 1 (xi>=0.04)']) disp(['Roll: T=', num2str(T_roll,'%.4f'), ' s -> Level 1
(T_R<=1.4s)'])  if real(lambda_spiral) < 0     disp('Spiral: stable -> Level 1') else
    T2s = log(2)/real(lambda_spiral);     disp(['Spiral: T2=', num2str(T2s,'%.2f'), ' s -> Level 1 (T2>=20s)']) end
disp(['DR: omega_n=', num2str(omega_n_DR,'%.4f'), ', xi=', num2str(zeta_DR,'%.4f'), ', xi*wn=',
num2str(zeta_DR*omega_n_DR,'%.4f'), ' -> Level 1'])  %%
========================================================================= %  12. SAS DESIGN - POLE PLACEMENT (A* = A -
B*K^T method) %% ========================================================================= disp(' ')
disp('============================') disp('   SAS DESIGN - POLE PLACEMENT') disp('============================')  %
--- PITCH DAMPER: Short Period xi 0.82 -> 0.95 --- xi_SP_new = 0.95; lam_sp_desired = omega_n_SP*(-xi_SP_new + 1j*sqrt(1-
xi_SP_new^2)); desired_poles_long = [lam_sp_desired, conj(lam_sp_desired), lambdaPH(1), lambdaPH(2)];  K_long =
place(ALong, BLong(:,1), desired_poles_long); ALong_SAS = ALong - BLong(:,1)*K_long;  disp(' '); disp('--- PITCH DAMPER --
-') disp(['Target: xi_SP = ', num2str(xi_SP_new)]) disp(['K_long = [', num2str(K_long(1),'%.4f'), ', ',
num2str(K_long(2),'%.4f'), ...       ', ', num2str(K_long(3),'%.4f'), ', ', num2str(K_long(4),'%.4f'), ']'])  eig_long_SAS
= eig(ALong_SAS); disp('Eigenvalues with SAS:'); disp(eig_long_SAS)  % --- YAW DAMPER: Dutch Roll xi 0.09 -> 0.20 ---
xi_DR_new = 0.20; lam_dr_desired = omega_n_DR*(-xi_DR_new + 1j*sqrt(1-xi_DR_new^2)); desired_poles_lat = [lam_dr_desired,
conj(lam_dr_desired), lambda_roll, lambda_spiral];  K_lat = place(ALat, BLat(:,1), desired_poles_lat); ALat_SAS = ALat -
BLat(:,1)*K_lat;  disp(' '); disp('--- YAW DAMPER ---') disp(['Target: xi_DR = ', num2str(xi_DR_new)]) disp(['K_lat = [',
num2str(K_lat(1),'%.4f'), ', ', num2str(K_lat(2),'%.4f'), ...       ', ', num2str(K_lat(3),'%.4f'), ', ',
num2str(K_lat(4),'%.4f'), ']'])  eig_lat_SAS = eig(ALat_SAS); disp('Eigenvalues with SAS:'); disp(eig_lat_SAS)  % ---
PITCH DAMPER COMPARISON FIGURE --- t_comp = linspace(0, 10, 1000); X0_long = [0; 1; 0; 0];  % initial Δw perturbation
[y_orig_long, ~] = initial(ss(ALong, zeros(4,1), eye(4), zeros(4,1)), X0_long, t_comp);
[y_sas_long, ~] = initial(ss(ALong_SAS, zeros(4,1), eye(4), zeros(4,1)), X0_long, t_comp);  fig =
figure('Visible','on','Position',[100 100 1000 750]); labels_long = {'\Delta u [m/s]','\Delta w [m/s]','\Delta q
[rad/s]','\Delta \theta [deg]'}; for k = 1:4     subplot(2,2,k);     d_orig = y_orig_long(:,k);     d_sas =
y_sas_long(:,k);     if k == 4  % theta in degrees         d_orig = rad2deg(d_orig);         d_sas = rad2deg(d_sas);
end     plot(t_comp, d_orig, 'b--', 'LineWidth', 1.8); hold on;     plot(t_comp, d_sas, 'r-', 'LineWidth', 1.8);
xlabel('Time [s]'); ylabel(labels_long{k}); grid on;     legend('Original', 'With SAS', 'Location', 'best');
set(gca,'FontSize',11); end sgtitle('SAS Comparison - Pitch Damper', 'FontSize', 14); exportgraphics(fig,
fullfile(outdir,'fig_SAS_PitchDamper.png'), 'Resolution', 200);  % --- YAW DAMPER COMPARISON FIGURE --- t_comp_lat =
linspace(0, 30, 2000); X0_lat = [1; 0; 0; 0];  % initial Δv perturbation (sideslip) [y_orig_lat, ~] = initial(ss(ALat,
zeros(4,1), eye(4), zeros(4,1)), X0_lat, t_comp_lat); [y_sas_lat, ~] = initial(ss(ALat_SAS, zeros(4,1), eye(4),
zeros(4,1)), X0_lat, t_comp_lat);  fig = figure('Visible','on','Position',[100 100 1000 750]); labels_lat = {'\Delta v
[m/s]','\Delta p [deg/s]','\Delta r [deg/s]','\Delta \phi [deg]'}; for k = 1:4     subplot(2,2,k);     d_orig =
y_orig_lat(:,k);     d_sas = y_sas_lat(:,k);     if k >= 2  % p, r, phi in degrees         d_orig = rad2deg(d_orig);
d_sas = rad2deg(d_sas);     end     plot(t_comp_lat, d_orig, 'b--', 'LineWidth', 1.8); hold on;     plot(t_comp_lat,
d_sas, 'r-', 'LineWidth', 1.8);     xlabel('Time [s]'); ylabel(labels_lat{k}); grid on;     legend('Original', 'With SAS',
'Location', 'best');     set(gca,'FontSize',11); end sgtitle('SAS Comparison - Yaw Damper', 'FontSize', 14);
exportgraphics(fig, fullfile(outdir,'fig_SAS_YawDamper.png'), 'Resolution', 200);  %%
========================================================================= %  13. BODE DIAGRAMS (numerical frequency
response) %% ========================================================================= I4 = eye(4);
w = logspace(-2, 2, 1200);  G_q_de = zeros(size(w)); G_theta_de = zeros(size(w)); G_p_da = zeros(size(w)); G_r_dr =
zeros(size(w));  for k = 1:length(w)     s = 1i*w(k);     GL = (s*I4 - ALong)\BLong;     GT = (s*I4 - ALat)\BLat;
G_q_de(k) = GL(3,1);     G_theta_de(k) = GL(4,1);     G_p_da(k) = GT(2,2);     G_r_dr(k) = GT(3,1); end  fig =
figure('Visible','on','Position',[100 100 700 500]); subplot(2,1,1) semilogx(w, 20*log10(abs(G_q_de)), 'LineWidth', 1.3);
grid on; ylabel('Magnitude (dB)'); title('Bode Diagram - q / \delta_e'); subplot(2,1,2) semilogx(w, angle(G_q_de)*180/pi,
'LineWidth', 1.3); grid on; xlabel('Frequency (rad/s)'); ylabel('Phase (deg)'); exportgraphics(fig,
fullfile(outdir,'fig_Bode_q_de.png'), 'Resolution', 200);  fig = figure('Visible','on','Position',[100 100 700 500]);
subplot(2,1,1) semilogx(w, 20*log10(abs(G_theta_de)), 'LineWidth', 1.3); grid on; ylabel('Magnitude (dB)'); title('Bode
Diagram - \theta / \delta_e'); subplot(2,1,2) semilogx(w, angle(G_theta_de)*180/pi, 'LineWidth', 1.3); grid on;
xlabel('Frequency (rad/s)'); ylabel('Phase (deg)'); exportgraphics(fig, fullfile(outdir,'fig_Bode_theta_de.png'),
'Resolution', 200);  fig = figure('Visible','on','Position',[100 100 700 500]); subplot(2,1,1) semilogx(w,
20*log10(abs(G_p_da)), 'LineWidth', 1.3); grid on; ylabel('Magnitude (dB)'); title('Bode Diagram - p / \delta_a');
subplot(2,1,2) semilogx(w, angle(G_p_da)*180/pi, 'LineWidth', 1.3); grid on; xlabel('Frequency (rad/s)'); ylabel('Phase
(deg)'); exportgraphics(fig, fullfile(outdir,'fig_Bode_p_da.png'), 'Resolution', 200);  fig =
figure('Visible','on','Position',[100 100 700 500]); subplot(2,1,1) semilogx(w, 20*log10(abs(G_r_dr)), 'LineWidth', 1.3);
grid on; ylabel('Magnitude (dB)'); title('Bode Diagram - r / \delta_r'); subplot(2,1,2) semilogx(w, angle(G_r_dr)*180/pi,
'LineWidth', 1.3); grid on;
xlabel('Frequency (rad/s)'); ylabel('Phase (deg)'); exportgraphics(fig, fullfile(outdir,'fig_Bode_r_dr.png'),
'Resolution', 200);  %% ========================================================================= %  14. ROOT LOCUS with S-
PLANE GRID (custom) %% ========================================================================= Kvec = linspace(0, 500,
700);  root_channels = {     ALong, BLong(:,1), [0 0 1 0], 'q/\delta_e',         [0 0.4470 0.7410];     ALong, BLong(:,1),
[0 0 0 1], '\theta/\delta_e',    [0.8500 0.3250 0.0980];     ALat,  BLat(:,2),  [0 1 0 0], 'p/\delta_a',         [0.9290
0.6940 0.1250];     ALat,  BLat(:,1),  [0 0 1 0], 'r/\delta_r',         [0.4940 0.1840 0.5560] };  fig =
figure('Visible','on','Position',[100 100 800 600], 'Color', 'w'); hold on; grid on; box on;  xlim_vals = [-15 2];
ylim_vals = [-10 10];  draw_sgrid_custom(xlim_vals, ylim_vals);  legend_entries = {'grid', '', '', ''};  for k =
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1:size(root_channels, 1)     A0 = root_channels{k,1};     B0 = root_channels{k,2};     Cfb = root_channels{k,3};     name
= root_channels{k,4};     col = root_channels{k,5};      pts_all = [];     for j = 1:length(Kvec)         K = Kvec(j);
Acl = A0 - B0*(K*Cfb);         rts = eig(Acl);         pts_all = [pts_all; rts(:).'];     end      for cidx =
1:size(pts_all, 2)         plot(real(pts_all(:,cidx)), imag(pts_all(:,cidx)), '.', 'Color', col, 'MarkerSize', 6);     end
open_poles = eig(A0);     plot(real(open_poles), imag(open_poles), 'x', 'Color', col, 'MarkerSize', 10, 'LineWidth', 2);
legend_entries{end+1} = name;     legend_entries{end+1} = ['open-loop poles ', name]; end
 xlabel('Real axis'); ylabel('Imaginary axis'); title('Root Locus - Main Aircraft Channels'); xlim(xlim_vals);
ylim(ylim_vals); legend(legend_entries, 'Location', 'bestoutside');  text(real(lambda_roll), 0.5, 'Roll mode', 'FontSize',
10); text(real(lambda_spiral)-0.8, 0.8, 'Spiral mode', 'FontSize', 10); text(real(lambdaDR_pair(1))+0.3,
imag(lambdaDR_pair(1))+0.5, 'Dutch roll', 'FontSize', 10);  exportgraphics(fig, fullfile(outdir,'fig_RootLocus.png'),
'Resolution', 200);  disp(' ') disp('=== DONE ===') disp(['Figures exported to: ', outdir])  %%
========================================================================= %  LOCAL FUNCTION: S-PLANE GRID %%
========================================================================= function draw_sgrid_custom(xlim_vals, ylim_vals)
hold on;     plot([xlim_vals(1) xlim_vals(2)], [0 0], 'k-', 'LineWidth', 0.8);     plot([0 0], [ylim_vals(1)
ylim_vals(2)], 'k-', 'LineWidth', 0.8);      zetas = [0.04 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 0.95 0.98];     rmax =
max(abs([xlim_vals ylim_vals]))*1.2;      for z = zetas         phi = acos(z);         x = linspace(0, rmax*cos(phi),
400);         y = tan(phi) * x;         plot(-x, y, ':', 'Color', [0.6 0.6 0.6], 'LineWidth', 0.8);         plot(-x, -y,
':', 'Color', [0.6 0.6 0.6], 'LineWidth', 0.8);          xt = -0.92*rmax*cos(phi);         yt = 0.92*rmax*sin(phi);
if xt >= xlim_vals(1) && xt <= xlim_vals(2) && yt >= ylim_vals(1) && yt <= ylim_vals(2)             text(xt, yt,
num2str(z), 'FontSize', 8, 'Color', [0.4 0.4 0.4]);             text(xt, -yt, num2str(z), 'FontSize', 8, 'Color', [0.4 0.4
0.4]);         end     end      wn_vals = [1 2 4 6 8 10 12 15];     th = linspace(pi/2, 3*pi/2, 500);     for wn = wn_vals
xc = wn*cos(th); yc = wn*sin(th);         plot(xc, yc, ':', 'Color', [0.75 0.75 0.75], 'LineWidth', 0.8);         if (-wn
>= xlim_vals(1) && -wn <= xlim_vals(2))             text(-wn, 0.35, num2str(wn), 'FontSize', 8, 'Color', [0.5 0.5 0.5],
'HorizontalAlignment', 'center');         end
    end end
```